



Published: October 2024

Guy Rolnik

Mary Donaldson

Investing in Elon Musk: The ESG Dilemma

It was late at night, just one month before the 2024 presidential election. Sarah McOwen, a senior analyst at a leading ESG advisory firm, was at her desk, deep in thought. Her computer screen displayed Tesla's financial history, a success story marked by revolutionary advancements in electric vehicles and clean energy. But next to it, a separate window revealed something more troubling: a feed filled with hundreds of tweets that Elon Musk had posted over the past two months, many stirring controversy.

As Sarah scrolled through the posts, she felt the weight of the decision she had to make. Her clients were eager for her recommendation on whether to invest in Musk's companies, and she knew her analysis would draw criticism from multiple angles. With its undeniable contributions to climate change mitigation, Tesla fits the ESG narrative on environmental impact. Yet Musk's recent acquisition and leadership of X (formerly Twitter) had ignited concerns about free speech, misinformation, and the degradation of civic discourse—issues that could undermine the very democratic fabric ESG investing sought to uphold.

She knew that whatever her final call, her boss would ask her to present her findings to the firm's investment committee. **The stakes were high.** The committee was split—half the board leaned liberal, half conservative—but there was one thing they all shared: a fear of looking foolish. They wouldn't tolerate a decision that could result in negative headlines in *The Wall Street Journal* or *The New York Times*.

The financial metrics for Tesla told one story of innovation and market dominance, but the political landscape Musk had stepped into with X raised more profound ethical and reputational questions.

Sarah took a deep breath. The decision was not just about balancing profits and values— it was about navigating the murky waters of politics, media scrutiny, and societal impact in a time when everything seemed hyper-charged by the approaching election. Whatever choice she made, it would be scrutinized—and she had to be prepared to defend it from all sides.

Early Life and Career

Elon Musk was born in June 1971 to parents Maye and Errol Musk in South Africa. From an early age, Musk showed an interest in technology and entrepreneurship. At only 12 years old, he taught himself computer programming and ultimately created his own video game, *Blastar*, which he sold for \$500¹. His parents ultimately divorced, and Musk ended up moving to Canada with his mother, where he attended Queen's University. After spending roughly two years at Queens University, Musk transferred to the University of Pennsylvania at roughly 21 years old; he graduated in 1997 with a dual degree in physics and economics².

After graduating from the University of Pennsylvania, Musk moved to California with the intention of earning a PhD in applied physics from Stanford University³. However, after spending only two days in the program, Musk decided to leave and instead focused on what he believed was an array of opportunities to disrupt the technology industry. Prior to being awarded his degree from the University of Pennsylvania, Musk, 24 years old at the time, co-created Zip2 with his brother. Zip2 was designed to help newspapers both manage and create local content and other directory services⁴. The premise of the business was to combine business listings, maps, and other editorial content, making it easier for readers to discover local services. By 1999, Compaq agreed to acquire Zip2 for roughly \$307 million⁵. Musk received approximately \$22 million from the sale of the business, making him a multi-millionaire at only 27 years old⁶. Musk ultimately utilized the proceeds from the sale of Zip2 to fund X.com.

Musk developed X.com with three other individuals, Ed Ho, Harris Fricker, and Christopher Payne⁷. The company was designed to operate as a robust financial services platform; specifically, Musk hoped X.com would become an all-in-one financial institution that could help customers with opening/maintaining checking accounts, provide brokerage services, originate loans, etc. The key feature, however, to X.com was its email-based payment system; users could send money to anyone who had an active email address. While this feature gained traction at a rapid pace, Musk faced competition from Confinity; Confinity's most popular product at the time, allowed for a secure transferring of funds via PalmPilot devices, while also offering email-based payments, like X.com⁸. When Musk was 29, in March 2000, X.com merged with Confinity; the

¹ Vendatu, *Elon Musk Biography*, <https://www.vedantu.com/biography/elon-musk>

² Ibid.

³ Ibid.

⁴ Ibid.

⁵ Britannica Money, *Zip2*, <https://www.britannica.com/money/Zip2>

⁶ CNN, *Elon Musk: How He Got Here and Where He's Going*, <https://www.cnn.com/interactive/2017/02/specials/elon-musk-timeline/elon-musk-timeline.html>

⁷ Encyclopedia.com, *X.com*, <https://www.encyclopedia.com/economics/encyclopedias-almanacs-transcripts-and-maps/xcom>

⁸ Mentorist, *The Founders: The Story of PayPal and the Entrepreneurs Who Shaped Silicon Valley*, <https://www.mentorist.app/books/the-founders-the-story-of-paypal-and-the-entrepreneurs-who-shaped-silicon->

combined business was later renamed PayPal by 2001. While Musk originally served as the CEO, he was later replaced by Peter Thiel due to internal disagreements pertaining to the direction of the business. In 2002, PayPal went public; later in the year, it was acquired by eBay for \$1.5 billion⁹. Musk was the largest shareholder at the time it was acquired, owning roughly 11.7% of PayPal's shares. As a result, Musk, 31 at the time of the sale, made about \$180 million from this deal¹⁰.

After the sale of PayPal, Musk again used the funds earned from this transaction to create new businesses. However, when it came to the creation of his new set of companies, such as SpaceX, Musk looked to put himself into positions where he would not be able to be fired or diluted out of control of the businesses. With a much deeper wallet than he had at the time of creating X.com, Musk used his financial standing to his advantage, spending massive sums of money to solidify his positioning within his next set of companies.

SpaceX, Tesla, and Solar City

SpaceX

Shortly after Musk benefitted from the sale of PayPal to eBay.com, Musk, 31, founded SpaceX in 2002, investing about \$100 million of his own money into the business. Musk created the business with the hopes of lowering the costs associated with space travel, and a longer-term goal of potentially enabling the human colonization of Mars. By 2008, SpaceX was the first privately funded company to send a liquid-fueled rocket, named the Falcon 1, into orbit¹¹. In 2012 the company's spacecraft, the Dragon, became the first commercial spacecraft to dock at the International Space Station¹², and by 2020, the company launched astronauts into space using its Crew Dragon capsule; the first private company to send astronauts into space.

SpaceX has also become an instrumental partner for NASA, through a variety of programs (including both a commercial cargo and crew program, alongside various exploration-focused programs). NASA granted SpaceX a \$1.6 billion contract to deliver cargo to the ISS. This original partnership has been extended multiple times¹³. By 2011, the partnership with NASA had expanded to cover not just the delivery of goods to space, but the transportation of humans as well. In 2011, SpaceX received approximately \$75

[valley/#:~:text=Confinity%20pursued%20mobile%20money%20transfer,was%20pursuing%20the%20wrong%20strategy.](#)

⁹ Wall Street Journal, *Elon Musk Buys Twitter. Here's How He Made His Fortune*, <https://www.wsj.com/story/how-elon-musk-made-the-fortune-hell-need-to-buy-twitter-c9e15543>

¹⁰ Ibid.

¹¹ Space.com, *SpaceX: Facts About Elon Musk's Private Spaceflight Company*, <https://www.space.com/18852-spacex-dragon.html>

¹² Space.com, *SpaceX Dragon: First Private Spacecraft to Reach the Space Station*, <https://www.space.com/18852-spacex-dragon.html>

¹³ Los Angeles Daily News, *SpaceX Awarded \$1.6 Billion Contract*, <https://www.dailynews.com/2008/12/24/spacex-awarded-16-billion-contract/>

million from the Commercial Crew Development program, followed by a \$440 million grant in 2012 via the Commercial Crew Integrated Capability Program¹⁴. By 2014, SpaceX was awarded approximately \$2.6 billion under NASA's Commercial Crew Transportation Capability to aid in the development and operation of crewmember transportation to the ISS¹⁵. In 2021, NASA announced the award of a \$2.9 billion contract to SpaceX, for the use of the company's Starship¹⁶. This was after three companies, including SpaceX, were granted contracts to develop landers that could transport humans from the Earth to the Moon's surface. In 2022, NASA awarded a contract modification to this original deal, worth an incremental \$1.5 billion, which SpaceX will use to further develop its Starship for lunar exploration with the culmination of the partnership occurring through a crewed lunar landing in 2027¹⁷.

The company is also responsible for the development of Starlink, a service that is designed to deliver high-speed broadband internet to locations where access has otherwise been either unavailable or inaccessible due to price/other issues. While Starlink was denied a nearly \$900 million rural broadband subsidy from the FCC's Rural Digital Opportunity Fund program in both 2022 and 2023, Starlink has benefitted from SpaceX's broader funding initiatives and NASA grants¹⁸.

Since its creation in 2002, SpaceX has gone through about 17 rounds of funding, raising roughly \$8.5 billion across the rounds¹⁹; this compares to the more than \$6 billion the company has received through its various partnerships with NASA and the U.S. Department of Defense. The company was recently valued by Bloomberg at more than \$200 billion²⁰. Through this time, SpaceX has rapidly scaled its profitability; from a net loss on about \$1.45 billion in revenue in 2019 to approximately \$3 billion in operating profits off \$9 billion in revenue²¹. This path to profitability aligns closely with the timing of Starlink's founding and scale; the service launched its first satellites into orbit in May of 2019 with the first beta test being made available to the public in November 2020. Sources indicate SpaceX will generate revenue of about \$15 billion in 2024, driven by

¹⁴ NASA, *NASA's Commercial Crew Program Progressing for Future of U.S. Human Spaceflight*, <https://www.nasa.gov/general/nasas-commercial-crew-program-progressing-for-future-of-u-s-human-spaceflight/>

¹⁵ CNN Business, *SpaceX Gets \$1.4 Billion Contract for Five More NASA Astronaut Launches*, <https://www.cnn.com/2022/08/31/tech/nasa-spacex-crew-launches-contract-extension-scn/index.html>

¹⁶ The New York Times, *SpaceX Wins NASA \$2.9 Billion Contract to Build Moon Lander*, <https://www.nytimes.com/2021/04/16/science/spacex-moon-nasa.html>

¹⁷ CNN, *SpaceX Gets \$1.4 Billion Contract for Five More NASA Astronaut Launches*, <https://www.cnn.com/2022/08/31/tech/nasa-spacex-crew-launches-contract-extension-scn/index.html>

¹⁸ FCC, *FCC Reaffirms Decision to Reject Starlink Application for Nearly \$900 Million in Subsidies*, <https://www.fcc.gov/document/fcc-reaffirms-decision-reject-starlink-application-nearly-900-million-subsidies>

¹⁹ Crunchbase, *Starlink*, https://www.crunchbase.com/organization/starlink-37f7/company_financials

²⁰ Bloomberg, *SpaceX Tender Offer Said to Value Company at Record \$210 Billion*, <https://www.bloomberg.com/news/articles/2024-06-27/spacex-tender-offer-said-to-value-company-at-record-210-billion>

²¹ Inc.com, *How Elon Musk Leaned on Starlink to Achieve Profitability at SpaceX*, <https://www.inc.com/chloe-aiello/how-elon-musk-leaned-on-starlink-to-achieve-profitability-at-spacex.html>

Starlink²². As the company is still private as of October 2024, there is no way to fully verify the financials of the company.

Tesla

Tesla was founded in 2003 by Marc Tarpenning and Martin Eberhard. It wasn't until 2004 that Musk became involved with Tesla after investing about \$6.5 million into the company's Series A²³. Musk became the largest investor in Tesla and chairman of the board after this investment. While Musk was not a founder of the company, he continued to invest heavily in Tesla and ultimately took over as CEO in 2008. Prior to becoming CEO, Musk released his "Master Plan" in 2006²⁴. Posted to Tesla's blog, Musk's note outlined his long-term vision and aspirations for the company's growth and overall strategy. Specifically, Musk's intention was to develop a sportscar (the Roadster) and use the money generated from the success of this car to invest in developing a more affordable car (the Model S), then ultimately leveraging the funds from this vehicle to build an even more affordable car. The plan also outlined Musk's intention to create zero-emission electric power generation options. About a decade later, Musk released the Master Plan, Part Deux²⁵. This second plan dove deeper into his thoughts on the company's future, with a focus on sustainability and autonomous driving capabilities. The Master Plan Part 3 was published on April 2023²⁶,

Musk has remained CEO of Tesla since taking on this role in 2008. While the company's market cap has been somewhat volatile in recent years, as of late October 2024, Tesla had a market cap of approximately \$682 billion²⁷. By comparison, Ford carried a valuation of approximately \$44 billion, while General Motors is valued at about \$58 billion²⁸. J.D. Power, an American data analytics company known for its reviews of automotive quality awarded Dodge the top spot in its U.S. initial quality study; Chevrolet ranked third, while Ford ranked ninth²⁹. Tesla received a score of 250, a score well below the lowest officially ranked automotive company but was not eligible to be officially ranked amongst the

²² Bloomberg, *SpaceX Eyes \$15 Billion in 2024 Sales on Starlink Strength*, <https://www.bloomberg.com/news/articles/2023-11-06/spacex-eyes-15-billion-in-sales-next-year-on-starlink-strength>

²³ The Street, *History of Tesla and Its Stock: Timeline, Facts, and Milestones*, <https://www.thestreet.com/technology/history-of-tesla-15088992#:~:text=2003%3A%20Tesla%20Motors%20is%20founded,%2C%20the%20all%2Delectric%20Roadster>

²⁴ Tesla, *The Secret Tesla Motors Master Plan (Just Between You and Me)*, <https://www.tesla.com/secret-master-plan>

²⁵ Tesla, *Master Plan, Part Deux*, <https://www.tesla.com/master-plan-part-deux>

²⁶ Tesla, *Master Plan Part 3*, https://www.tesla.com/ns_videos/Tesla-Master-Plan-Part-3.pdf

²⁷ Factset, as of October 2024.

²⁸ Ibid.

²⁹ J.D. Power, *New-Vehicle Quality Mainly Dependent on Trouble-Free Technology, J.D. Power Finds*, <https://www.jdpower.com/sites/default/files/file/2020-06/2020070%20U.S.%20IQS.pdf>

other brands as it does not meet ranking criteria³⁰. Specifically, Tesla did not allow J.D. Power survey owners across 15 states, as is required.

Autonomous Vehicle Initiative

Musk's plan with Tesla was to build an automobile with extensive technological capabilities. Starting in 2014, Musk was quoted on CNN stating "autonomous cars will definitely be a reality"³¹. The same year, Tesla prefaced how it saw the future of its technology in a press release, stating: "Imagine having your car check your calendar in the morning, calculate travel time to your first appointment based on real-time traffic data, automatically open the garage door with Homelink, carefully back out of a tight garage, and pull up to your door ready for your commute. Of course, it could also warm or cool your car to your preferences and select your favorite morning news stream"³². Tesla's Autopilot was sent out to select Tesla vehicles in 2014, with the full rollout to Model S drivers in 2015. At the time, Musk shared with Fortune magazine, "I think we have all the pieces [for fully autonomous driving capabilities], it's just about refining those pieces, putting them in place, and making sure they work across a huge number of environments – then we're done"³³. In the same article, Musk was quoted as stating that the problem of developing autonomous driving software is "a much easier problem than people think it is"³⁴.

Tesla continued the rollout of its vehicles. By 2016, Musk indicated that "all Tesla vehicles leaving the factory have all the hardware necessary for Level 5 autonomy"³⁵. While in that same year Tesla released a demonstration video which showed a Model X driving itself, in an interview with Insider, Musk stated that 2017 would be the year Tesla launched a fully autonomous vehicle for public consumption. He stated that the demonstration would showcase "full autonomy all the way from L.A. to New York...and then have the car go and park itself". Musk also tweeted that "in ~2 years, summon should work anywhere connected by land & not blocked by borders, e.g., you're in LA and the car is in NY"³⁶. By the end of 2017, Musk addressed a conference on Neural Information Processing Systems and stated that self-driving cars were now just two years away; a little more than two years after he had originally stated the technology would be publicly available³⁷.

³⁰ Ibid.

³¹ U.S. News, *Will Tesla's 'D' Be A Self-Driving Car?*, <https://www.usnews.com/news/articles/2014/10/09/will-teslas-d-be-a-self-driving-car>

³² Electrek, *Understanding Tesla's Self-Driving Features: The Autopilot*, <https://electrek.co/2015/06/16/understanding-teslas-self-driving-features-the-autopilot/>

³³ Fortune, *Elon Musk Says Tesla Vehicles Will Drive Themselves in Two Years*, <https://fortune.com/2015/12/21/elon-musk-interview/>

³⁴ Ibid.

³⁵ The New York Times, *Inside Tesla as Elon Musk Pushed an Unflinching Vision for Self-Driving Cars*, <https://www.nytimes.com/2021/12/06/technology/tesla-autopilot-elon-musk.html>

³⁶ X.com, Elon Musk

³⁷ Teslarati, *Tesla is Developing Its Own Custom AI Chip, says Elon Musk*, <https://www.teslarati.com/tesla-developing-custom-ai-chip-says-elon-musk/>

In 2018, Musk doubled down on his belief in Tesla's technological advantage over its competition. When speaking with Kara Swisher, editor-at-large of Recode, Musk said "I don't want to sound overconfident, but I would be very surprised if any of the car companies exceeded Tesla in self-driving, in getting to full self-driving. They're just not good at software, and this is a software problem"³⁸. The following year, Musk instead emphasized Tesla's upgradable software; he tweeted that "buying a car in 2019 that can't upgrade to full self-driving is like buying a horse instead of a car in 1919"³⁹. Over the next few years, Musk continued to emphasize Tesla's track toward developing fully autonomous driving technology, culminating in the recent Cybercab announcement. Musk estimates that the vehicles, which will be available for \$30,000 and completely autonomous, will enter production by 2026⁴⁰.

While Musk and Tesla have been eyeing an eventual launch of the autonomous vehicle technology, many other players have entered the race. Most notable is Waymo, which has launched fully autonomous vehicles in a variety of U.S. cities, which consumers are able to use today⁴¹. Despite Tesla's evolution in its technology, its current tech stack is incomparable to Waymo, as one of the only initiatives to have launched a fully autonomous vehicle for public consumption.

These initiatives have not been without failures. In 2018, an Uber self-driving vehicle struck and killed a woman pushing her bicycle across a highway; the vehicle's perception system was not able to properly classify the woman, and the safety driver was distracted, watching a television show on her cell phone⁴². The following year, in 2019, a fatal crash occurred when a Tesla Model S that was set on autopilot crashed into a tractor-trailer, immediately killing the driver.⁴³ Waymo is not immune to these challenges; the National Highway Traffic Safety Administration has launched an investigation into 22 different incidents involving Waymo cars⁴⁴.

³⁸ Forbes, *Tesla Could Have Full Self-Driving Cars On The Road By 2019, Elon Musk Says*, <https://www.forbes.com/sites/jeanbaptiste/2018/11/07/tesla-could-have-full-self-driving-cars-on-the-road-by-2019-elon-musk-says/>

³⁹ X.com, Elon Musk

⁴⁰ Teslarati, *Tesla Cybercab Production to Reach 2M a Year*, <https://www.teslarati.com/tesla-cybercab-2-million-production/>

⁴¹ Waymo, *Meet Waymo One*, <https://waymo.com/#:~:text=Operating%20in%20multiple%20cities,Angeles%20and%20Austin%20coming%20soon.>

⁴² CNN Business, *Uber Self-Driving Car Test Driver Pleads Guilty to Endangerment in Pedestrian Death Case*, <https://www.cnn.com/2023/07/29/business/uber-self-driving-car-death-guilty/index.html>

⁴³ CBS News, *Tesla Autopilot Not Responsible for Fatal 2019 Crash in California, Jury Finds in Landmark Case*, <https://www.cbsnews.com/news/tesla-autopilot-not-responsible-fatal-2019-crash-california-jury-finds-landmark-case/>

⁴⁴ Spectrum News One, *NHTSA Is Investigating 22 Incidents With Waymo Self-Driving Cars*, <https://spectrumnews1.com/ca/southern-california/transportation/2024/05/14/nhtsa-is-investigating-22-incidents-with-waymo-self-driving-cars#:~:text=The%20agency%20said%20it%20is,violation%20of%20traffic%20safety%20laws.>

SEC Investigation

After tweeting earlier in 2018 that Musk had secured funding to take Tesla private, he later indicated that the funding he was referencing in this tweet was from Saudi Arabia's Public Investment Fund. Per Musk, the fund was interested in assisting in taking Tesla private, but at the time Musk tweeted, it had not made any formal agreement to purchase Tesla.

The SEC later launched an investigation into whether Musk had potentially violated securities laws by making misleading and false statements that could have impacted Tesla's stock price. After Musk's tweet went public, Tesla shares rose roughly 11% during the trading day, resulting in a loss of roughly \$1.3 billion by short sellers in mark-to-market losses⁴⁵. By September 2018, the SEC filed a lawsuit against Musk⁴⁶, accusing him of fraud and citing that the tweet he sent was misleading to the public (as a deal had not been finalized). While Musk's intention was to fight the charges, he ultimately settled with the SEC in October 2018, where both he and Tesla were ordered to pay fines of \$20 million each⁴⁷. As of the end of October 2018, Tesla's market cap was about \$56 billion⁴⁸.

SolarCity

SolarCity was founded in 2006 by Peter and Lyndon Rive, two of Musk's cousins.⁴⁹ From the time the company was founded, Musk has served as its chairperson; he personally invested about \$10 million during the company's early stages. From Musk's perspective, he consistently viewed SolarCity as a key player in the larger goal of developing a sustainable energy ecosystem. The company was founded with the goal of making solar power more accessible and affordable to both residential and commercial customers through leasing and financing opportunities.

In 2016, Tesla acquired SolarCity for roughly \$2.6 billion. Certain Tesla shareholders believed that SolarCity was a troubled business and viewed the deal as opportunistic for Musk and his family, as they owned a significant share of the company at the time it was acquired by Tesla. At the time of its acquisition, SolarCity had never turned a profit; the company posted a loss of roughly \$710 million in 2015. It also carried a substantial

⁴⁵ CNBC, *Elon Musk's Tweet About Going Private Costs Tesla Short Sellers \$1.3 Billion*,

<https://www.cnn.com/2018/08/07/elon-musks-tweet-about-going-private-costs-tesla-short-sellers-more-t.html>

⁴⁶ U.S. Securities and Exchange Commission Enforcement and Litigation, SEC v. Elon Musk Case No.18-cv-8865, [https://www.sec.gov/enforcement-litigation/distributions-for-harmed-investors/sec-v-elon-musk-case-no-18-cv-8865-sdny-sec-v-tesla-inc-case-no-18-cv-8947-sdny#:~:text=18%2Dcv%2D8947%20\(S.D.N.Y.\),-Oct.&text=On%20September%2027%2C%202018%2C%20the,a%20publicly%20traded%20company%2C%20private.](https://www.sec.gov/enforcement-litigation/distributions-for-harmed-investors/sec-v-elon-musk-case-no-18-cv-8865-sdny-sec-v-tesla-inc-case-no-18-cv-8947-sdny#:~:text=18%2Dcv%2D8947%20(S.D.N.Y.),-Oct.&text=On%20September%2027%2C%202018%2C%20the,a%20publicly%20traded%20company%2C%20private.)

⁴⁷ PBS, *Elon Musk Ordered to Abide By SEC Settlement Over 2018 Tweets*,

<https://www.pbs.org/newshour/economy/elon-musk-ordered-to-abide-by-sec-settlement-over-2018-tweets#:~:text=The%20whole%20dispute%20stems%20from,pre%2Dapproval%20of%20his%20tweets.>

⁴⁸ Companies Market Cap, *Tesla*, <https://companiesmarketcap.com/tesla/marketcap/>

⁴⁹ Crunchbase, *SolarCity*, <https://www.crunchbase.com/organization/solarcity>

amount of debt – over \$3 billion when Musk agreed to acquire it⁵⁰. Musk stated that the acquisition would aid Tesla in its long-term mission of driving more sustainable energy sources. Musk was later sued by shareholders who alleged Musk engaged in a “bailout” of SolarCity, and coerced the board into the acquisition of a company that shareholders believed was worth nowhere near what Tesla paid. Ultimately, Musk was found not liable; had he lost, he would have had to pay out nearly \$2 billion⁵¹.

The Strategic Management of International Relationships

When Tesla opened its factory in Shanghai in 2019, the company’s first factory outside of the U.S., Musk was able to secure favorable terms with the Chinese government. Most notable was the ability of Tesla to maintain complete ownership of its Chinese factory without a local partner, something that had historically been nearly impossible⁵². Tesla also benefitted from various land subsidies and tax incentives. Musk has intentionally avoided making any direct criticisms against the Chinese government, or speaking out negatively about certain political topics, such as human rights challenges. Instead, Musk remains hyper-focused on maintaining strong ties with the Chinese government, aligning Tesla’s mission with China’s aim to dominate the global EV market.

Musk’s relationship with Prime Minister Narendra Modi of India has also been of great importance. Musk himself met with Modi to discuss Tesla’s possible entry into the Indian car market, as Modi’s government had expressed an interest in promoting both renewable energy and electric mobility options. Modi has been excited by the prospect of foreign investment and manufacturing, under the broader “Make in India” initiative.

Musk has responded stating that he’d be willing to open a local manufacturing facility in the country, if they were to commit to more favorable tax incentives and local manufacturing requirements for Tesla.

Musk’s Run-Ins with the Mainstream Media and the SEC

Media Backlash

Musk has had several high-profile confrontations with large media outlets throughout his career; most notably with The New York Times and The Economist between 2018 and 2019.

⁵⁰ Reuters, *Court Upholds Musk’s Win in \$13 Billion Lawsuit Over Tesla-SolarCity Deal*, <https://www.reuters.com/legal/court-upholds-ruling-musk-over-tesla-solarcity-deal-2023-06-06/>

⁵¹ CNBC, *Elon Musk Wins Shareholder Lawsuit Over Tesla’s \$2.6 Billion SolarCity Acquisition*, <https://www.cnbc.com/2022/04/27/elon-musk-wins-shareholder-lawsuit-over-the-companys-2point6-billion-solarcity-acquisition.html#:~:text=Had%20he%20lost%2C%20Musk%20could,Vice%20Chancellor%20Joseph%20R.>

⁵² AP News, *Tesla Moves Forward With a Plan to Build an Energy-Storage Battery Factory in China*, <https://apnews.com/article/china-tesla-megapack-battery-factory-shanghai-306b4f6678c6827080a7bf49d55d12ee>

In 2018, Tesla was facing challenges with the production of its Model 3 vehicle. The Model 3 was intended to be Tesla's mainstream vehicle, driving significant market demand at its affordable price point. However, many questioned whether Tesla would be able to achieve its production targets and ultimately reach profitability.

In April of that year, The Economist released an article questioning the firm's solvency. After Elon tweeted, "We are sad to report that Tesla has gone completely and totally bankrupt," on April 1st, an attempt at an April Fool's joke, the Economist wrote that his tweet "is uncomfortably close to the truth". The article continued speaking out against Musk and Tesla's failed timelines and promises, stating, "Tesla has repeatedly failed to meet its own targets; [Musk] vowed to produce 2,500 Model 3s a week by the end of March, rising to 5,000 a week by the end of June. Despite superhuman efforts by workers and managers ([Musk] is personally supervising the production of the new model and claims to be sleeping at the factory), on April 3rd, Tesla confirmed that it is producing only around 2,000 Model 3 saloons a week. Expectations were so low among analysts and investors that Tesla's flagging share price rebounded after that announcement"⁵³. Beyond the Economists' claims around Tesla's failure to meet its own timelines, the article homed in on the company's possible solvency issue. After citing ongoing investor concerns, it stated "On March 27th, Moody's, a credit-rating agency, downgraded Tesla's debt, cautioning the firm 'will likely need to raise additional capital during the second half of 2019'. Jefferies, a bank, predicts that Tesla will need \$2.5 billion to \$3 billion this year"⁵⁴. Musk was quick to refute the Economists' statements, tweeting that the company was not in any position to need to raise money and would soon be profitable (Exhibit 2).

A few months after The Economist article was published, Musk posted on Twitter that he had been considering taking Tesla private and had secured the funding to do so (Exhibit 3). A few weeks later, The New York Times published an article summarizing an interview conducted with Musk. Per the source, Musk emphasized the stress and exhaustion he was facing as the CEO of Tesla and indicated that 2018 had been "the most difficult and painful year⁵⁵" of his entire career. The article cited Musk's somewhat erratic behavior with the interviewer, sharing that during the interview, "Mr. Musk alternated between laughter and tears. He said he had been working up to 120 hours a week recently...and said he had not taken more than a week off since 2001, when he was bedridden with malaria⁵⁶". Elaborating on Musk's take-private tweet, the article stated that the tweet(s) "were the most recent of several flare-ups that had drawn scrutiny. He wrangled with short-sellers and belittled analysts for asking 'boring, boneheaded' questions. And after sending a team of engineers from one of his companies to help rescue

⁵³ The Economist, *Tesla Is Heading For a Cash Crunch*, <https://www.economist.com/business/2018/04/05/tesla-is-heading-for-a-cash-crunch>

⁵⁴ Ibid.

⁵⁵ The New York Times, *Elon Musk Details 'Excruciating' Personal Toll of Tesla Turmoil*, <https://www.nytimes.com/2018/08/16/business/elon-musk-interview-tesla.html>

⁵⁶ Ibid.

members of a stranded soccer team, he lashed out at a cave diver who was dismissive of the gesture, deriding him on Twitter as a ‘pedo guy’⁵⁷.

Like his reaction against the Economist, Musk pushed back on The New York Times’ narrative, stating that he believed the media had dramatized his struggles. On August 28, 2018, Musk tweeted “for the record, my voice cracked once during the NY Times article. That’s it. There were no tears”. Despite this, Musk still found himself embroiled in chaos pertaining to his recent tweet. Beyond the mainstream media, equity analysts questioned Musk’s thought process, and questioned whether Musk was attempting to manipulate the market. Musk was later fined by the SEC; both he and Tesla were ordered to pay \$20 million, with Musk also having to relinquish his role on Tesla’s board.

Musk also found himself fighting against other sources, such as Business Insider and Reuters. In 2018, an article written by Business Insider highlighted a variety of struggles both Tesla and Musk faced during the year, titled “Tesla is experiencing a painful year – here’s everything that has gone wrong so far”⁵⁸. The article itself is 20+ pages in length, and closes citing a Wall Street Journal report that indicated the “DOJ is investigating whether Tesla misled investors about Model 3 production”⁵⁹. Musk questioned its validity; tweeting earlier this year “never believe anything written by Business Insider. That fake publication is not worth reading at all. Utter trash”.

The media pressure continued into 2020, when Musk fought the media about his response to the COVID-19 pandemic. Musk disagreed with COVID-driven lockdowns and restrictions; he subsequently reopened a Tesla factory in California earlier than permitted. At the time of the factory’s reopening, Musk tweeted “Tesla is restarting production today against Alameda County rules. I will be on the line with everyone else. If anyone is arrested, I ask that it only be me”⁶⁰. He followed up by tweeting “Also, all other auto companies in the US are approved to resume. Only Tesla has been singled out. This is super messed up!”⁶¹.

The Purchase of Twitter and its Restructuring Under Musk’s Leadership

The Purchase

In mid-April 2022, Musk made an unsolicited offer to take Twitter private for \$44 billion, equating to about \$54.20 per share⁶². Musk financed the deal through investments by

⁵⁷ Ibid.

⁵⁸ Business Insider, *Tesla Is Experiencing A Painful Year – Here’s Everything That Has Gone Wrong So Far*, <https://www.businessinsider.com/tesla-challenges-in-2018-2018-4>

⁵⁹ Ibid.

⁶⁰ Fox News, *Elon Musk Reopens Tesla Factory Closed By Coronavirus, ‘If Anyone is Arrested I Ask That It Only Be Me’*, <https://www.foxnews.com/auto/tesla-elon-musk-factory-reopens-california-coronavirus-tweet>

⁶¹ Ibid.

⁶² The Street, *Twitter to X Corp: A Complete Timeline of Elon Musk’s Twitter Takeover*, <https://www.thestreet.com/technology/twitter-to-x-the-complete-timeline-of-elon-musks-twitter-purchase#:~:text=Elon%20Musk's%20net%20worth%20was,billion%20acquisition%20of%20the%20platform.>

individuals, venture capital firms, banks, and his own personal money. The breakdown of the \$44 billion offer is as follows⁶³:

- **Prince Alwaleed Bin Talal al Saud (Saudi Prince)** – he agreed to convert his Twitter shares, worth about \$2 billion, into stake in the new private enterprise.
- **The Qatar Investment Authority** – the fund which has roughly \$450 billion of capital, agreed to invest \$375 million in the deal.
- **Binance** – shortly after Musk publicized his interest in taking Twitter private, the company reached out to Musk directly and contributed \$500 million towards the acquisition.
- **Andreessen Horowitz** – one of the most famous venture capital groups operating in the Silicon Valley gave Musk about \$400 million for the acquisition.
- **Sequoia Capital** – like Andreessen Horowitz, Sequoia is a notable investment firm in the technology sector; it provided Musk with about \$800 million in funding.
- **Larry Ellison** – the co-founder and chairman of Oracle; provided Musk with about \$1 billion.
- **Jack Dorsey** – one of Twitter’s co-founder and the former CEO agreed to roll over his investment in Twitter to Musk. At the time, his investment was valued at about \$1 billion.
- **Bank of America, Barclays, and Morgan Stanley** – the combination of the three banks provided Musk with about \$13 billion in funding, taken on as debt.

At the time Musk made the offer, Twitter’s shares were trading at \$45.85; Musk’s offer represented a 20% premium to the shares’ value at the time it was made⁶⁴. While Twitter’s board of directors debated utilizing a poison pill strategy to block the purchase, it ultimately accepted Musk’s offer on April 25, 2022⁶⁵.

On May 13, 2022, Musk decided to put his offer on hold, tweeting: “Twitter deal temporarily on hold pending details supporting calculation that spam/fake accounts do indeed represent less than 5% of users”⁶⁶. Many questioned whether bots were the true reason Musk pushed back against his offer; from the time Musk first tweeted about acquiring the company on April 4, 2022 to May 13, 2022, the NASDAQ had declined roughly 19%, from \$14,532.55 to \$11,805.00⁶⁷.

By early July, Musk tried to stop the deal entirely, arguing that Twitter had not been forthcoming with information about the number of spam/bot accounts that existed on

⁶³ Washington Post, *Here’s Who Helped Elon Musk Buy Twitter*, <https://www.washingtonpost.com/technology/2022/12/24/elon-musk-twitter-funders/>

⁶⁴ Ibid.

⁶⁵ Ibid.

⁶⁶ X.com, Elon Musk

⁶⁷ NASDAQ Historical Prices, https://www.nasdaq.com/market-activity/index/comp/historical?page=7&rows_per_page=100&timeline=y5

the platform. A few days after Musk tried to halt the deal, Twitter sued Musk to force the closure of the acquisition, believing that Musk did not have sufficient data to remove himself from a binding deal. Of note, at the time Musk originally tried to pause the deal, the stock had traded down to about \$40.72 per share; by early July, the stock was trading at less than \$40 per share⁶⁸. While Musk reiterated that his concerns around his purchase of the company were due to the bots on the platform, others have speculated that Musk was aiming to get a better purchase price on the company, given the dip in the stock price in the months after his initial offer was made.

The Restructuring

After having a court date set for October 2022, Musk ultimately decided to go through with the purchase of the company, closing on Twitter on October 27, 2022 for his original price of \$44 billion. The same day the acquisition was closed, Musk **fired** various key executives (including the CEO, CFO, and CLO). Musk himself assumed the role of CEO after this dismissal. In addition to the removal of these executives, he also reduced Twitter's global workforce by roughly 50%, cutting the employee base down to about 3,700 from 7,500. In many cases, the layoffs were said to have disproportionately impacted people whose responsibility it was to prevent hateful speech/other prohibited content from impacting the platform⁶⁹.

To diversify the company's revenue streams (as prior to the acquisition, it was reliant on advertising revenues), Musk introduced a subscription model, named Twitter Blue. Users could pay a nominal amount per month, to purchase the blue verification mark that had historically only been made available to celebrities, journalists, and other notable figures. Twitter Blue faced problems almost immediately; many bad actors attempted to impersonate brands, institutions, and other famous figures. To rectify the issue, Twitter ultimately paused the service and relaunched it with different colored checkmarks for different account types. Despite the rebrand, Twitter remained predominantly reliant on advertising to generate revenues.

Many advertisers temporarily paused spending on the platform immediately after Musk's takeover. Specifically, Twitter was said to have lost about half of its top 100 advertisers post takeover by Musk. Estimates indicated that these advertisers had spent nearly \$2 billion on Twitter from 2020 to their pause in 2022, nearly half of that amount being spent in 2022 alone prior to walking away from the platform⁷⁰. Across the 50

⁶⁸ Ibid.

⁶⁹ NBC News, *Days Before the Midterms, Twitter Lays Off Employees Who Fight Misinformation*, <https://www.nbcnews.com/tech/misinformation/twitter-fires-employees-fight-misinformation-midterm-elections-rcna55750>

⁷⁰ NPR, *Twitter Has Lost 50 of its Top 100 Advertisers Since Elon Musk Took Over, Report Says*, <https://www.npr.org/2022/11/25/1139180002/twitter-loses-50-top-advertisers-elon-musk#:~:text=Twitter%20has%20lost%2050%20of,Musk%20took%20over%2C%20report%20says&text=via%20Getty%20Images->

remaining advertisers from Twitter's top 100 list, seven were said to have slowed their advertising spend on the platform; these seven advertisers had paid Twitter about \$255 million from 2020 to late 2022⁷¹. Advertisers, such as Mondelez International, indicated the **pullback** in advertising spend was due to "concerns about hate speech on the platform"⁷². Advertiser spend has not since returned in full force to the platform. In 2022, Twitter generated about \$4.4 billion in revenue, largely comprised of advertising revenues. In 2023, estimates and internal documents point to total revenue for X of \$3.4 billion; revenue is expected to decline again in 2024 to about \$2.9 billion⁷³.

Advertisers commonly look for brand safe platforms when deploying advertising dollars. This means that advertisers have comfort in knowing that when a platform is able to determine where exactly their advertisement is placed, it will be placed alongside/before/after content that the brand knows will not be offensive or harmful to any group of individuals. After Musk's takeover, Musk had fired or been served with resignations from roughly 75% of the company's staff, including the loss of the bulk of the company's trust and safety teams⁷⁴; as a result, the content on X began to shift. Specifically, "pre-Musk takeover, the bulk of the conversation on [trending political narratives, such as the civil war, election fraud, citizen policing of voting, and allegations of pedophilia/grooming], was focused on refuting misinformation, deriding the toxic speech, and mocking those who spread it"⁷⁵, per a report by Digital Planet, a research initiative out of Tufts University. "Post-Musk takeover, the quality of the conversation has decayed, with more extremists and purveyors of hateful content testing the boundaries of what Twitter might allow"⁷⁶. According to Brandwatch, a social media analysis tool, in the first three months of Musk's tenure at X, the rate of daily tweets "containing slurs against Black Americans more than tripled...from October through March, tweets referring to the LGBTQ+ community alongside slurs such as 'groomer' rose 119%...reports of harassment on Twitter rose 6% this year, according to the ADL"⁷⁷.

Despite the increase in hate speech, Musk tweeted that the platform "would not become a 'free-for-all hellscape'"⁷⁸. The company also stated that its focus was on limiting the

.Half%20of%20Twitter's%20top%20100%20advertisers%20appear%20to,be%20advertising%20on%20the%20web site.

⁷¹ Ibid.

⁷² Media Matters, *In Less Than a Month, Elon Musk Has Driven Away Half of Twitter's Top 100 Advertisers*, <https://www.mediamatters.org/elon-musk/less-month-elon-musk-has-driven-away-half-twitters-top-100-advertisers>

⁷³ Social Media Today, *Data Suggests X Could Post a Significant Loss in 2024*.

<https://www.socialmediatoday.com/news/x-formerly-twitter-could-post-a-significant-loss-in-2024/730370/>

⁷⁴ Time, *Harmful Content Has Surged on Twitter, Keeping Advertisers Away*, <https://time.com/6295711/twitters-hate-content-advertisers/>

⁷⁵ Fortune, *The Quality of Twitter Conversation Has 'Decayed' After Elon Musk's Takeover*, <https://fortune.com/2022/11/08/elon-musk-twitter-conversation-quality-decayed/>

⁷⁶ Ibid.

⁷⁷ Time, *Harmful Content Has Surged on Twitter, Keeping Advertisers Away*, <https://time.com/6295711/twitters-hate-content-advertisers/>

⁷⁸ Fortune, *The Quality of Twitter Conversation Has 'Decayed' After Elon Musk's Takeover*, <https://fortune.com/2022/11/08/elon-musk-twitter-conversation-quality-decayed/>

number of people that viewed the hate speech, not necessarily the volume of content. X indicated that impressions of hate speech content were 30% lower on average than before Musk's takeover⁷⁹.

The Business of Social Media

An Overview on the Social Media Landscape

Social media platforms have permeated globally, with remarkable user engagement across platforms. Facebook has more than 3 billion monthly active users, while YouTube, WhatsApp, and Instagram all have more than 2 billion monthly active users⁸⁰. TikTok, the newest entrant to the social media sector, has nearly 1.7 billion monthly active users, with X at nearly 600 million⁸¹. Recent estimates indicate that users spend roughly 2 hours and 23 minutes per day on social media, across apps that they utilize⁸². Opinions on the impact of social media are mixed. A 2019 study indicated that "routine social media use, like responding to content others share daily, is actually linked to positive social well-being and overall mental health"⁸³. Mark Zuckerberg has long boasted what he sees as the benefits of social media, providing people with a place to be actively engaged with others and build relationships⁸⁴. Others, such as the Columbia Mailman School of Public Health, called out the negative impact of social media, stating: "...Social media platforms are important sources of socialization and relationship-building for many young people...social media can also provide platforms for bullying and exclusion, unrealistic expectations about body image and sources of popularity, normalization of risk-taking behaviors, and can be detrimental to mental health"⁸⁵.

Business Model

Social media companies, including X, Facebook, Instagram, WhatsApp, Pinterest, Snapchat, etc. rely primarily on revenues from advertisements to grow their businesses. Since 2021, Meta has generated more than \$100 billion in advertising revenues annually and is one of the world's highest valued public companies⁸⁶. While some of these

⁷⁹ Time, *Harmful Content Has Surged on Twitter, Keeping Advertisers Away*, <https://time.com/6295711/twitters-hate-content-advertisers/>

⁸⁰ Exploding Topics, *Top 35 Social Media Platforms (September 2024)*, <https://explodingtopics.com/blog/top-social-media-platforms>

⁸¹ Ibid.

⁸² SOAX, *What Is The Average Time Spend on Social Media Each Day?*, <https://soax.com/research/time-spent-on-social-media#:~:text=Research%20highlights%3A%20On%20average%2C%20people,spent%20scrolling%20through%20different%20platforms.>

⁸³ Sage, *Association of Social Media Use with Social Well-Being, Positive Mental Health, and Self-Rated Health: Disentangling Routine Use from Emotional Connection to Use*, <https://journals.sagepub.com/doi/full/10.1177/1090198119863768>

⁸⁴ CNBC, *Mark Zuckerberg Doesn't Like Your Scrolling Habits: Social Media is For 'Building Relationships', Not Just Consuming Content*, <https://www.cnbc.com/2022/09/04/zuckerberg-social-media-is-for-building-relationships-not-scrolling.html>

⁸⁵ Columbia Mailman School of Public Health, *Just How Harmful Is Social Media? Our Experts Weigh-In*, <https://www.publichealth.columbia.edu/news/just-how-harmful-social-media-our-experts-weigh>

⁸⁶ Factset Estimates, October 2024.

companies have alternative revenue streams (as is the case with Meta's Reality Labs revenue segment), the primary revenue generator for these businesses is still spending from advertising agencies or small businesses. The way the advertisements are delivered tends to differ by platform; in the case of Facebook and Instagram, users are served ads via the newsfeed, stories, reels, and videos. While Snapchat offers advertisements through "Snap Ads" (videos shown between stories) and "sponsored lenses", which act as branded camera filters.

Most social media companies also utilize the use of real-time bidding, an auction format that allows advertisers to purchase ad space on the platform. Advertisers are competing in real-time in the auction to secure a place on the platform; the platform will then use all available data to determine who the advertisement should be shown to generate the best ROAS (return on advertising spend) for the advertiser.

For advertisers, the ability to advertise on social media platforms can have many benefits. Typically, advertisements can be bucketed into two broad categories: brand and performance. Brand advertising means that a brand is simply trying to educate the public about the offerings of their brand, while performance advertising is used to try to get a potential customer to do something (whether that's click on an advertisement, redirect themselves to the company's page, etc.). The social media platforms offer a medium for advertisers looking to allocate budgets for both aforementioned purposes; beyond that, social media platforms, through their vast datasets, tend to excel at targeting. This means that advertisers may have to spend more to advertise on the platforms, however, there's more confidence for the advertisers that their advertisements will reach their desired audiences. With more than 5 billion global social media user identities, there's no audience that cannot be reached through social media marketing.

The Algorithm

When these platforms were in their nascent stages of operation, the algorithms across most apps were set in chronological order, with only a few advertisements interspersed throughout content from people that you were "friends" with or followed. As the platforms developed, so did the algorithm; content from friends/followed pages were still prioritized within the news feed, but rather than showing posts in chronological order, posts that had a high level of engagement (through likes, comments, shares, etc.) were given priority in the feed. This was done to show users content they were most likely to interact with first, continuing to promote the platforms' goal of user interaction. As time went on, the algorithm continued to shift – moving away from prioritizing content from friends or existing pages that the user followed, to promoting content that the algorithm believed the user would want to interact with. Oftentimes, this included promoting sponsored content into users' news feeds.

Meta carried the advantage of its extensive database of user data. Due to its data trove, it could "develop models that learned to infer the existence not only of broad categories

like ‘women’ and ‘men’, but of very fine-grained categories like ‘women between 25 and 34 who liked Facebook pages related to yoga’ and [target] ads to them. The finer-grained the targeting, the better chance of a click, which would give advertisers more bang for their buck”⁸⁷. At Meta, there is no singular team that is responsible for ranking content; engineers develop and modify the machine-learning technologies to train the models based on their own teams’ objectives. Different teams tend to have differing objectives; the machine learning system at the same time has grown increasingly complex, making it impossible for anyone to manage all the different components. This has meant for Meta that the company’s core process for quality control is through measurement and experimentation practices.

In 2021, Frances Haugen, a previous employee of Meta, shared tens of thousands of internal documents with both the SEC and Congress. The documents were then utilized by the Wall Street Journal, which published an article about how Meta “knew about how it contributed to harms ranging from its impact on teens’ mental health and the extent of misinformation on its platforms, to human traffickers’ open use of its services”⁸⁸. The documents provided by Haugen “paint a picture of a company that is often aware of the harms to which it contributes – but is either unwilling or unable to act against them”⁸⁹. Specifically, Hagen cited what she saw as fundamental flaws in the company’s algorithm. In her testimony to the Senate, she stated: “Facebook...knows – they have admitted in public – that engagement-based ranking is dangerous without integrity and security systems, but then not rolled out those integrity and security systems in most of the languages in the world...It is pulling families apart. An in places like Ethiopia it is literally fanning ethnic violence”⁹⁰.

Meta rejected Haugen’s arguments, citing its annual spend of about \$5 billion to keep its platform safe⁹¹.

Regulatory and Legal Concerns

Section 230 is considered “the 26 words that created the internet”⁹² and the foundation of the business model for social networks. The section was enacted as part of the

⁸⁷ MIT Technology Review, *The Facebook Whistleblower Says its Algorithms Are Dangerous. Here’s Why.*, <https://www.technologyreview.com/2021/10/05/1036519/facebook-whistleblower-frances-haugen-algorithms/>

⁸⁸ Time, *Frances Haugen Whistleblower Profile*, <https://time.com/6121931/frances-haugen-facebook-whistleblower-profile/>

⁸⁹ Ibid.

⁹⁰ MIT Technology Review, *The Facebook Whistleblower Says its Algorithms Are Dangerous. Here’s Why.*, <https://www.technologyreview.com/2021/10/05/1036519/facebook-whistleblower-frances-haugen-algorithms/>

⁹¹ Time, *Frances Haugen Whistleblower Profile*, <https://time.com/6121931/frances-haugen-facebook-whistleblower-profile/>

⁹² CNN Business, *These 26 Words ‘Created the Internet’. Now the Supreme Court May be Coming for Them*, <https://www.cnn.com/2023/02/18/tech/section-230-explainer/index.html>

Communications Decency Act (CDA), which was passed into law in 1996⁹³. Per the Electronic Frontier Foundation, “the free and open internet as we know it couldn’t exist without Section 230”⁹⁴. Under this law, social media platforms, and other websites/platforms are not to be made legally liable for any user-generated content. In laymen’s terms, this implies that the social media platforms are not the publishers of any user content, and therefore cannot be sued for any harmful or possibly defamatory post that makes it onto their platform, unlike a traditional newspaper or publisher. Beyond this, Section 230 also provides platforms with the right to “act in good faith”⁹⁵ in their monitoring of content. Platforms, based upon their own judgment, can remove certain content from their platform at any given time, without sacrificing any legal protection provided via this law. By doing this, Section 230 was intended to promote the monitoring and removal of any hate speech or other harmful content, without the platforms having to worry about a lawsuit over being too restrictive/not restrictive enough.

Through Section 230, this allowed platforms to encourage free speech from all sides of an argument, without having to worry about taking on liability as a platform in allowing these differing viewpoints to be posted. Many have referred to Section 230 as a Good Samaritan-type statute; these normally limit liability for individuals who have no relationship to a person that is in trouble/jeopardy, and therefore does not have to provide any assistance⁹⁶. Questions have arisen as to whether Section 230 has accomplished the goal of preventing harm to users as was intended; traditionally, businesses have a duty of care which urges them to avoid reasonably causing harm to a customer and taking necessary steps to continuously prevent potential harm to their customers. However, as Section 230 exists, the traditional duty that would be placed on the business to protect its customers from harm, in this case, the assumption that all hate speech or other harmful commentary must be banished from platforms, does not necessarily exist⁹⁷.

While Section 230 has existed since 1996, the law has found itself at the center of many court cases.

In 2017, an attack was carried out in a Turkish nightclub, killing 39 people. Plaintiffs, all relatives of victims of the attack, sued Google, Twitter, and Facebook, arguing that the platforms provided “substantial assistance to ISIS” by allowing the group to utilize their

⁹³ Congressional Research Service, *Section 230: A Brief Overview*, <https://crsreports.congress.gov/product/pdf/IF/IF12584#:~:text=Restricting%20Access%20to-Objectionable%20Material,to%20be%20obscene%2C%20.%20.%20.>

⁹⁴ Electronic Frontier Foundation, *Section 230*, <https://www.eff.org/issues/cda230#:~:text=Section%20230%20Protects%20Us%20All&text=Important%20court%20rulings%20on%20Section,videos%20that%20others%20find%20objectionable.>

⁹⁵ Congressional Research Service, *Section 230: A Brief Overview*, <https://crsreports.congress.gov/product/pdf/IF/IF12584#:~:text=Restricting%20Access%20to-Objectionable%20Material,to%20be%20obscene%2C%20.%20.%20.>

⁹⁶ Congress.gov Meeting Notes, *Why Section 230 Isn’t Really a Good Samaritan Provision*, <https://www.congress.gov/117/meeting/house/111407/documents/HHRG-117-IF16-20210325-SD013.pdf>

⁹⁷ Ibid.

social media platforms⁹⁸. The district court dismissed the case, stating that there was not a direct relationship between the platforms' actions and the terrorist attack⁹⁹, while the Ninth Circuit Court of Appeals reversed the decision, allowing the case to proceed¹⁰⁰. In mid-May 2023, the Supreme Court issued a unanimous opinion rejecting the plaintiffs' claims – specifically stating that “providing a generally available internet service does not equate to ‘aiding and abetting’ terrorism under the ATA”¹⁰¹. Further, the opinion clarified that “platforms cannot be held liable for user-generated content unless there is a concrete nexus between their conduct and specific criminal acts”¹⁰². This ruling reaffirmed the protections for social media companies under Section 230.

In another case pertaining to the 2015 ISIS in France, which killed 130 people, including 23-year-old U.S. citizen Nohemi Gonzalez, Gonzalez's family ultimately sued Google, citing the company's direct and secondary liability due to ISIS's use of YouTube¹⁰³. Specifically, the plaintiff argued Google was directly liable under the Anti-Terrorism Act and indicated that Google “aided and abetted ISIS by allowing the terrorist group to use YouTube”¹⁰⁴. After the district court dismissed the case, the Ninth Circuit Court of Appeals affirmed the dismissal, referring to Section 230 of the Communications Decency Act¹⁰⁵. In May of 2023, the Supreme Court issued a unanimous opinion, which avoided a direct addressing of Section 230, but vacated the Ninth Circuit's judgement, recommending a reconsideration of the case, despite its recent ruling in the *Twitter v. Taamneh* case¹⁰⁶.

In August 2024, the U.S. Court of Appeals (for the Third Circuit) cited that Section 230 does not protect TikTok from liability for its algorithm-driven recommendations, specifically as it relates to the death of ten-year-old Nylah Anderson. Anderson passed away after attempting the “Blackout Challenge”, which was recommended to her via TikTok's algorithm. In its ruling, the court cited that it believes TikTok's algorithm was its own “expressive activity”, as it curates and recommends content to users; as a result, it would not be covered under Section 230 as it stands today. TikTok had responded to the ruling requesting a rehearing; the fate of Section 230 going forward remains unknown.

⁹⁸ Harvard Law Review, *Twitter, Inc. v. Taamneh*, <https://harvardlawreview.org/print/vol-137/twitter-inc-v-taamneh/>

⁹⁹ *Ibid.*

¹⁰⁰ *Ibid.*

¹⁰¹ *Ibid.*

¹⁰² *Ibid.*

¹⁰³ Justia – U.S. Supreme Court, *Gonzalez v. Google LLC*, 598 U.S. __ (2023), <https://supreme.justia.com/cases/federal/us/598/21-1333/>

¹⁰⁴ *Ibid.*

¹⁰⁵ *Ibid.*

¹⁰⁶ *Ibid.*

The Campaign for “Absolute Free Speech” on X

Musk envisions X as a digital town square, reliant on free speech and the communication of various viewpoints. However, studies reveal that social media platforms often amplify extreme viewpoints and misinformation due to algorithmic designs that prioritize engagement over balanced discourse. This dynamic leads to polarization and echo chambers rather than fostering open, diverse dialogue in the way a traditional town square might.

Musk has long positioned himself as a champion for the right to free speech. In April 2022, Musk tweeted “free speech is the bedrock of a functioning democracy, and Twitter is the digital town square where matters vital to the future of humanity are debated”¹⁰⁷. He had previously expressed a disdain for content moderation practices and argued that in many cases the viewpoints of certain groups, specifically those that are politically conservative in America, had been silenced because of heightened content monitoring. More recently in August of 2024, he tweeted: “freedom of speech is the bedrock of democracy. If the truth is suppressed, it is impossible to make an informed voting decision. The degree to which freedom of speech is being undermined around the world is incredibly alarming”¹⁰⁸. Musk portrays X as a platform where power is decentralized, giving every user an equal voice. Nevertheless, the research on these issues shows a more complicated picture. Power on X is often concentrated among influencers and high-profile users, whose content is amplified by algorithms designed to maximize engagement. This results in unequal visibility, where a few voices dominate the conversation while others remain marginalized.

On August 12, 2024, Musk hosted a live chat with Donald Trump that was streamed to users via X. The stream faced technical challenges, and pushback from sources like the BBC, which stated that in the two-hour interview, “Donald Trump made a number of questionable and false claims – which largely went unchallenged”¹⁰⁹. Trump cited the biggest threat to the globe today as nuclear war, rather than global warming, and later stated that global warming may present a great real estate opportunity as it will create more beachfront properties. Where platforms like Facebook may have otherwise caveated the discussion with a fact-checking option, the stream was broadcast to more than a million users without any content moderation barriers. Off X, Musk himself has become one of Trump’s most prominent allies, attending rallies and supporting the former president on X. Musk has posted on X regarding “doubts about election security issues”, while becoming one of the most “viral voices” in this years’ election¹¹⁰. Leading up to the presidential election in 2024, “immigration and voter fraud [had] become

¹⁰⁷ X.com, Elon Musk

¹⁰⁸ Ibid.

¹⁰⁹ BBC, *Trump’s Chat with Musk on X Fact-Checked*, <https://www.bbc.com/news/articles/c0jpn2q76n1o>

¹¹⁰ CBS News, *The X Factor: How Trump Ally Elon Musk is Using Social Media to Prime Voter Mistrust Ahead of 2024 Election*, <https://www.cbsnews.com/news/elon-musk-trump-social-media-election-2024/>

Musk's most frequently posted and engaged with policy topic, garnering about 10 billion views. Musk posted more than 1,300 times about the topic overall"¹¹¹.

At the time of the acquisition, Musk indicated he wanted to remove the existing content moderation policies and establish new, clearer rules; this included allowing certain banned members to rejoin the Twitter community. In abolishing the prior content moderation and fact-checking policies, misinformation has been able to bypass traditional media gatekeepers, at the expense of public trust in the news.

Incidents of Musk's Tweets Resulting in Regulatory/Government Push Back

Musk's Fight Against the Brazilian Supreme Court

In April 2024, X shared that a Brazilian court had ordered the platform to block an undisclosed number of profiles; if the platform refused to do so, the Brazilian court indicated it would levy heavy fines against the corporation. In response, Elon Musk stated that he would reduce restrictions on Brazilian accounts that were previously suspended and threatened to shut down X in Brazil. In response, Alexandre de Moraes, one of Brazil's 11 Supreme Court judges, opened an inquiry into Musk for obstruction of justice. Musk responded by stating that censorship within Brazil is worse than "any other country in the world in which [X] operates", he called Moraes a dictator, and implied that de Moraes should be both impeached and placed "on trial for his crimes".

By August 2024, Brazil had moved to suspend access to X, after the company missed a court-imposed deadline to name a legal representative in Brazil. In response, Musk posted to X, stating “they’re shutting down the #1 source of truth in Brazil”. In addition to the banning of the platform, de Moraes looked to enforce a fine of roughly \$8,900 per day for any person or company that utilized a VPN to access X during its ban. At the time this occurred, it was estimated that roughly 40 million Brazilians, equating to about 18% of the country’s population, accessed X at least one per month, per eMarketer. The blocking of the platform was met with mixed opinions, pertaining to the legitimacy of the ban of the platform in Brazil. Despite the potential fines, many Brazilian users still inquired about the use of VPNs and how that would continue to enable them to utilize X during the ban. Further, the Brazilian Bar Association pushed back on de Moraes fines, arguing that sanctions should not be imposed prior to “ensuring an adversarial process and the right to full defense.”

¹¹¹ Bloomberg, *Elon Musk is Now X's Biggest Promoter of Anti-Immigrant Conspiracies*, https://www.bloomberg.com/graphics/2024-musk-x-election-influence-immigration/?accessToken=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzb3VyY2UiOiJTdWJzY3JpYmVyb2lmZGVkQXJ0aWNsZSIsImhhdCI6MTcyOTc3NjM1OCwiZXhwIjoxNzMwMzgxMTU0LCJhcnRpY2xlSWQ1IiwiaWF0Ij01OTY0fHJHRzAwMCI6ImJjb25uZWN0SWQ1IiwiaXBkeE40TQ2RTQ5MzE0RUVBQjcwM0NDQzU0MkQ4ODE1MSJ9.hhXgVuBLH87875Jl2EO10Xq0EbdGr2_YmJxHMBJ2UbK

By the end of September 2024, X had complied with orders from the Brazilian Supreme Court, including blocking select accounts from the platform, identifying a legal representative in Brazil, and agreeing to pay fines imposed for prior noncompliance, according to a source familiar with X's filing. Despite this, it's still too soon to tell whether the ongoing dispute between X, Musk, and de Moraes is really settled.

X and the Global Business Empire of Elon Musk

After taking over, Musk published Twitter's recommendation algorithm to GitHub; Musk thought it would be beneficial for the public was able to see exactly how content was curated on the platform. Musk later came under fire for questions around the algorithm's tweaking to feature certain content. Musk himself later admitted that his tweets were being shown more prominently in users' feeds – he later blamed this on an “internal system bug”. Musk also came under fire for accusations of altering his own follower count, after it surged to an unusually high-level post acquisition. He later denied these claims, stating that the increase was due to an increase in attention to his platform as the new owner/CEO.

Musk has also used X to express his views on regulatory issues. He has made numerous statements on Twitter that have been interpreted as a benefit that a government could reap if they were to accommodate Musk's requests. Some examples include:

1. **Transferring of Tesla's Headquarters:** During the height of the COVID-19 pandemic driven lockdowns, Musk took to X to publicly threaten to move Tesla's headquarters out of California, stating: “Frankly, this is the final straw. Tesla will now move its HQ and future programs to Texas/Nevada immediately.¹¹²” By 2021, Tesla's headquarters had been moved to Austin, Texas.
2. **Musk vs. Sanders:** Musk has engaged in numerous public spats with U.S. representatives, often pertaining to taxes. After Senator Bernie Sanders called on high earners to pay higher taxes, Musk shot back calling out his plans to sell off a large portion of his stock holdings, and quipped that he had forgotten Sanders was still alive¹¹³. Musk doubled down on his disdain for the public debates over taxing the wealthy, citing the negative consequences higher taxes would have on his business and stock performance.
3. **Complaints Over European Regulations:** Musk's commentary on government practices and policies does not only pertain to the U.S. While the construction of Giga Berlin was underway in Germany, Musk tweeted, citing the extended bureaucratic processes and other environmental regulations that had delayed the process¹¹⁴. He suggested that German regulators streamline the process of approval for businesses, in this case, Tesla, as he believed the inclusion of these businesses would benefit Germany's overall economy.

¹¹² X.com, Elon Musk

¹¹³ Ibid.

¹¹⁴ Ibid.

4. **Dogecoin and Other Crypto Thoughts:** Beyond sharing memes and other commentary on the cryptocurrency market, Musk has often called for less strict regulations for cryptocurrencies¹¹⁵; specifically, for those like Dogecoin and Bitcoin where Musk himself or one of his companies was an owner.

The Old vs. New Algorithm

Prior to Musk's acquisition, Twitter provided users with two options for their timeline: a chronological view and algorithmic view. Under the chronological view, tweets would appear in a user's timeline in real-time order, while the algorithmic view curated content into the users' feed based on what the platform assumed Twitter would find most interesting. Twitter's algorithm would promote and prioritize tweets with high engagement metrics, including the number of likes, number of retweets, number of comments, etc., to determine what content should be promoted. Beyond this, the platform also relied on the use of hashtags and other trending topics; users were able to review the real-time most used hashtags to capture sentiment around real-time events. This allowed users to organically discover new topics and information based on trending conversations.

After Musk completed his acquisition of Twitter, the algorithm changed. Under his leadership, the new platform, X, became more reliant on the "for you" page. This page is algorithmically curated based on the interests of the user, measured by monitoring how the user interacts and engages with specific content across the platform. As a result, the page will often show tweets from people that users are not currently following, but might be interested in, and more prominent advertiser content. There were also allegations made that Musk modified the algorithm to prioritize content from his account – with many users finding tweets from Musk across the platform; this became especially prevalent after an incident in 2023 after the Super Bowl where a tweet from Musk performed worse than a tweet from President Joe Biden. It was after this incident that Musk had engineers modify the algorithm to prioritize his page and content.

Musk and AI

In December 2015, Musk, alongside Sam Altman, unveiled OpenAI at a weeklong AI conference. By mid-2016, OpenAI was releasing its first batch of AI-enabled software, "a toolkit for building artificially intelligent systems by a way of technology called 'reinforcement learning'"¹¹⁶. The company was pitched as a "billion-dollar effort to push AI as far as it will go"¹¹⁷. Many at the time, including Musk, saw massive opportunities through this new technology, across a variety of use cases. At the time Musk announced this partnership, Google and Facebook had already been utilizing AI-powered tools through deep learning initiatives, as well as using AI to improve efficiencies and the

¹¹⁵ Ibid.

¹¹⁶ Wired, *Inside OpenAI, Elon Musk's Wild Plan to Set Artificial Intelligence Free*, <https://www.wired.com/2016/04/openai-elon-musk-sam-altman-plan-to-set-artificial-intelligence-free/>

¹¹⁷ Ibid.

consumer experience on their platforms. For Tesla specifically, Musk acknowledged the need of AI to power self-driving cars, alongside the need of the technology to support his SpaceX initiatives. Despite these references, Musk did at the time cite the risk associated with humans losing control of systems that were powerful enough to learn without human interference¹¹⁸. Specifically, Musk referenced the utilization of AI as “summoning the demon”¹¹⁹; from his point of view, he worried that AI could surpass human capabilities and intelligence in ways that humans would not be able to control. Beyond that, he believed that AI may end up being harmful to society to the extent that it could act autonomously. Musk has advocated for regulation around AI, “[urging] California to pass the AI bill requiring tech companies to conduct safety testing on their AI models. Musk...has long advocated for AI regulation, likening it to rules for any technology that could pose risks to the public. He specifically called for the passage of California’s SB 1047 bill to address these concerns”¹²⁰.

Despite his concerns, Musk remained involved in OpenAI. In 2017, OpenAI shifted to become a for-profit entity. Musk then decided he “wanted majority equity, initial board control, and to be CEO”¹²¹. The existing team and other investors in OpenAI could not reach an agreement with Musk on his position as CEO and other requests; Musk then pitched the opportunity for OpenAI to merge with Tesla, stating that this move was “exactly right...Tesla is the only path that could even hope to hold a candle to Google. Even then, the probability of being a counterweight to Google is small. It just isn’t zero”¹²². Ultimately, by 2018, Musk departed OpenAI, stating to the team that the probability of success in the business was zero¹²³. At the time, he also alleged that he would take an opportunity to attempt to build a competing technology by himself.

In the time since departing OpenAI, Musk has maintained a largely negative view on chatbots and other AI-powered large language models. In 2023, Musk signed an open letter which called for a six-month pause on AI-experiments, citing extreme risks to society and humanity¹²⁴. Despite this, Musk still sees an opportunity to utilize the power of AI, citing his desire to create “a maximum truth-seeking AI that tries to understand the nature of the universe...I’m worried about the fact that [ChatGPT] is being trained to be politically correct, which is another way of saying untruthful things”¹²⁵. Musk has also

¹¹⁸ Ibid.

¹¹⁹ The Washington Post, *Elon Musk: ‘With Artificial Intelligence We Are Summoning the Demon*, <https://www.washingtonpost.com/news/innovations/wp/2014/10/24/elon-musk-with-artificial-intelligence-we-are-summoning-the-demon/>

¹²⁰ DigWatch, *Elon Musk Pushes for AI Safety Law in California*, <https://dig.watch/updates/elon-musk-pushes-for-ai-safety-law-in-california>

¹²¹ OpenAI, *OpenAI and Elon Musk*, <https://openai.com/index/openai-elon-musk/>

¹²² Ibid.

¹²³ Ibid.

¹²⁴ CBS News, *Elon Musk Says He Wants to Create His Own “Safer” Version of ChatGPT called TruthGPT*, <https://www.cbsnews.com/news/elon-musk-artificial-intelligence-chatgpt-truthgpt-safer-ai-tucker-carlson-fox-news-interview/>

¹²⁵ Ibid.

cited the need for the continued development of AI, specifically as it pertains to the needs of both Tesla and SpaceX.

The Impact of AI on X

Prior to Musk's acquisition, Twitter used AI to moderate content on its platform. However, as Twitter became X, Musk relaxed many of the content moderation practices in his advocacy for free speech on the platform, becoming less reliant on AI within this facet of the business. Despite this, Musk has continued to utilize AI in bolstering X's algorithm and personalization of the platform. AI is foundational to X's content recommendation algorithm, and will continue to be a necessity for the algorithm going forward. There is speculation that Musk may use the technology developed by xAI to strengthen X's algorithm, but Musk has not yet integrated the two entities.

Where Does this Leave Musk and X Today?

The White House had denounced Musk in 2023 for the **promotion** of what they noted was "**antisemitic and racist hate**"¹²⁶, while others have called out Musk's push for "free speech" as a veil for his interest in causing controversy and allowing for inappropriate conversations. Despite all of this, the U.S. government remains heavily dependent on Musk and his corporations. The domestic government had agreed to up to \$1.2 billion of SpaceX launches in 2024, after paying in the tens of millions for a new secure communications system back in 2023¹²⁷.

It's yet to be seen what will happen to X and Musk going forward, with a contentious election nearing in America. While many public figures, such as Mark Zuckerberg, have tried to pull away from taking sides in political conversations, Musk has continued to lean heavily into the political debate – oftentimes freely sharing his thoughts on candidates' policies via X.

The future of X also remains in flux, with Fidelity recently citing a valuation of \$9.4 billion for the company; a substantial decline from the \$44 billion Musk paid to acquire the service¹²⁸. Under Musk's ownership, advertisers have expressed concerns about the content that is now being made available (and sometimes promoted) on the platform; advertisers don't want to run the risk of their brand being associated with extreme viewpoints or other potentially offensive content. Some have speculated that X's user base has declined since Musk's acquisition of the platform, however, as the company has gone private, there's no clear way to validate this assumption.

¹²⁶ Variety, *White House Slams Elon Musk's 'Abhorrent Promotion of Antisemitic and Racist Hate'*, <https://variety.com/2023/digital/news/white-house-condemns-elon-musk-antisemitic-racist-hate-1235795049/>

¹²⁷ The Information, *Inside SpaceX's Lucrative New Government Satellite Program*, <https://www.theinformation.com/articles/inside-spacexs-lucrative-new-government-satellite-program>

¹²⁸ Forbes, *Elon Musk's X Is Now Worth Around A Fifth of the \$44 Billion He Paid For It, Fidelity Says*, <https://www.forbes.com/sites/tylerroush/2024/09/30/elon-musks-x-is-now-worth-around-a-fifth-of-the-44-billion-he-paid-for-it-fidelity-says/#:~:text=Key%20Facts,of%20August%2C%20the%20company%20said.>

Overall, its yet to be seen what impact Musk's push for free speech may ultimately have on X, his other companies, and social media more broadly going forward.

Exhibit 1: Tesla Master Plans

The Secret Tesla Motors Master Plan (just between you and me)

Elon Musk, Co-Founder & CEO of Tesla Motors, August 2, 2006

As you know, the initial product of Tesla Motors is a high performance electric sports car called the Tesla Roadster. However, some readers may not be aware of the fact that our long term plan is to build a wide range of models, including affordably priced family cars. This is because the overarching purpose of Tesla Motors (and the reason I am funding the company) is to help expedite the move from a mine-and-burn hydrocarbon economy towards a solar electric economy, which I believe to be the primary, but not exclusive, sustainable solution.

Critical to making that happen is an electric car without compromises, which is why the Tesla Roadster is designed to beat a gasoline sports car like a Porsche or Ferrari in a head to head showdown. Then, over and above that fact, it has twice the energy efficiency of a Prius. Even so, some may question whether this actually does any good for the world. Are we really in need of another high performance sports car? Will it actually make a difference to global carbon emissions?

Well, the answers are no and not much. However, that misses the point, unless you understand the secret master plan alluded to above. Almost any new technology initially has high unit cost before it can be optimized and this is no less true for electric cars. The strategy of Tesla is to enter at the high end of the market, where customers are prepared to pay a premium, and then drive down market as fast as possible to higher unit volume and lower prices with each successive model.

Without giving away too much, I can say that the second model will be a sporty four door family car at roughly half the \$89k price point of the Tesla Roadster and the third model will be even more affordable. In keeping with a fast growing technology company, all free cash flow is plowed back into R&D to drive down the costs and bring the follow on products to market as fast as possible. When someone buys the Tesla Roadster sports car, they are actually helping pay for development of the low cost family car.

Now I'd like to address two repeated arguments against electric vehicles — battery disposal and power plant emissions. The answer to the first is short and simple, the second requires a bit of math:

Batteries that are not toxic to the environment!

I wouldn't recommend them as a dessert topping, but the Tesla Motors Lithium-Ion cells are not classified as hazardous and are landfill safe. However, dumping them in the trash would be throwing money away, since the battery pack can be sold to recycling companies (unsubsidized) at the end of its greater than 100,000-mile design life. Moreover, the battery isn't dead at that point, it just has less range.

Power Plant Emissions aka "The Long Tailpipe"

A common rebuttal to electric vehicles as a solution to carbon emissions is that they simply transfer the CO₂ emissions to the power plant. The obvious counter is that one can develop grid electric power from a variety of means, many of which, like hydro, wind, geothermal, nuclear, solar, etc. involve no CO₂ emissions. However, let's assume for the moment that the electricity is generated from a hydrocarbon source like natural gas, the most popular fuel for new US power plants in recent years.

The H-System Combined Cycle Generator from General Electric is 60% efficient in turning natural gas into electricity. "Combined Cycle" is where the natural gas is burned to generate electricity and then the waste heat is used to create steam that powers a second generator. Natural gas recovery is 97.5% efficient, processing is also 97.5% efficient and then transmission efficiency over the electric grid is 92% on average. This gives us a well-to-electric-outlet efficiency of $97.5\% \times 97.5\% \times 60\% \times 92\% = 52.5\%$.

Despite a body shape, tires and gearing aimed at high performance rather than peak efficiency, the Tesla Roadster requires 0.4 MJ per kilometer or, stated another way, will travel 2.53 km per mega-joule of electricity. The full cycle charge and discharge efficiency of the Tesla Roadster is 86%, which means that for every 100 MJ of electricity used to charge the battery, about 86 MJ reaches the motor.

Bringing the math together, we get the final figure of merit of $2.53 \text{ km/MJ} \times 86\% \times 52.5\% = 1.14 \text{ km/MJ}$. Let's compare that to the Prius and a few other options normally considered energy efficient.

The fully considered well-to-wheel efficiency of a gasoline powered car is equal to the energy content of gasoline (34.3 MJ/liter) minus the refinement & transportation losses (18.3%), multiplied by the miles per gallon or km per liter. The Prius at an EPA rated 55 mpg therefore has an energy efficiency of 0.56 km/MJ. This is actually an excellent number compared with a "normal" car like the Toyota Camry at 0.28 km/MJ.

Note the term hybrid as applied to cars currently on the road is a misnomer. They are really just gasoline powered cars with a little battery assistance and, unless you are one of the handful who have an aftermarket hack, the little battery has to be charged from the gasoline engine. Therefore, they can be considered simply as slightly more efficient gasoline powered cars. If the EPA certified mileage is 55 mpg, then it is indistinguishable from a non-hybrid that achieves 55 mpg. As a friend of mine says, a world 100% full of Prius drivers is still 100% addicted to oil.

The CO₂ content of any given source fuel is well understood. Natural gas is 14.4 grams of carbon per mega-joule and oil is 19.9 grams of carbon per mega-joule. Applying those carbon content levels to the vehicle efficiencies, including as a reference the Honda combusted natural gas and Honda fuel cell natural gas vehicles, the hands down winner is pure electric:

Car	Energy Source	CO ₂ Content	Efficiency	CO ₂ Emission
Honda CNG	Natural Gas	14.4 g/MJ	0.32 km/MJ	45.0 g/km
Honda FCX	Nat Gas-Fuel Cell	14.4 g/MJ	0.35 km/MJ	41.1 g/km
Toyota Prius	Oil	19.9 g/MJ	0.56 km/MJ	35.8 g/km
Tesla Roadster	Nat Gas-Electric	14.4 g/MJ	1.14 km/MJ	12.6 g/km

The Tesla Roadster still wins by a hefty margin if you assume the average CO₂ per joule of US power production. The higher CO₂ content of coal compared to natural gas is offset by the negligible CO₂ content of hydro, nuclear, geothermal, wind, solar, etc. The exact power production mixture varies from one part of the country to another and is changing over time, so natural gas is used here as a fixed yardstick.

Becoming Energy Positive

I should mention that Tesla Motors will be co-marketing sustainable energy products from other companies along with the car. For example, among other choices, we will be offering a modestly sized and priced solar panel from SolarCity, a photovoltaics company (where I am also the principal financier). This system can be installed on your roof in an out of the way location, because of its small size, or set up as a carport and will generate about 50 miles per day of electricity.

If you travel less than 350 miles per week, you will therefore be “energy positive” with respect to your personal transportation. This is a step beyond conserving or even nullifying your use of energy for transport – you will actually be putting more energy back into the system than you consume in transportation! **So, in short, the master plan is:**

1. Build sports car
2. Use that money to build an affordable car
3. Use *that* money to build an even more affordable car
4. While doing above, also provide zero emission electric power generation options

Don't tell anyone.

Master Plan, Part Deux

Elon Musk, July 20, 2016

The first master plan that I wrote 10 years ago is now in the final stages of completion. It wasn't all that complicated and basically consisted of:

1. Create a low volume car, which would necessarily be expensive
2. Use that money to develop a medium volume car at a lower price
3. Use *that* money to create an affordable, high volume car

And...

4. Provide solar power. No kidding, this has literally been on our website for 10 years.

The reason we had to start off with step 1 was that it was all I could afford to do with what I made from PayPal. I thought our chances of success were so low that I didn't want to risk anyone's funds in the beginning but my own. The list of successful car company startups is short. As of 2016, the number of American car companies that haven't gone bankrupt is a grand total of two: Ford and Tesla. Starting a car company is idiotic and an electric car company is idiocy squared.

Also, a low volume car means a much smaller, simpler factory, albeit with most things done by hand. Without economies of scale, anything we built would be expensive, whether it was an economy sedan or a sports car. While at least some people would be prepared to pay a high price for a sports car, no one was going to pay \$100k for an electric Honda Civic, no matter how cool it looked.

Part of the reason I wrote the first master plan was to defend against the inevitable attacks Tesla would face accusing us of just caring about making cars for rich people, implying that we felt there was a shortage of sports car companies or some other bizarre rationale. Unfortunately, the blog didn't stop countless attack articles on exactly these grounds, so it pretty much completely failed that objective.

However, the main reason was to explain how our actions fit into a larger picture, so that they would seem less random. The point of all this was, and remains, accelerating the advent of sustainable energy, so that we can imagine far into the future and life is still good. That's what "sustainable" means. It's not some silly, hippy thing -- it matters for everyone.

By definition, we must at some point achieve a sustainable energy economy or we will run out of fossil fuels to burn and civilization will collapse. Given that we must get off fossil fuels anyway and that virtually all scientists agree that dramatically increasing atmospheric and oceanic carbon levels is insane, the faster we achieve sustainability, the better.

Here is what we plan to do to make that day come sooner:

Integrate Energy Generation and Storage

Create a smoothly integrated and beautiful solar-roof-with-battery product that just works, empowering the individual as their own utility, and then scale that throughout the world. One ordering experience, one installation, one service contact, one phone app.

We can't do this well if Tesla and SolarCity are different companies, which is why we need to combine and break down the barriers inherent to being separate companies. That they are separate at all, despite similar origins and pursuit of the same overarching goal of sustainable energy, is largely an accident of history. Now that Tesla is ready to scale Powerwall and SolarCity is ready to provide highly differentiated solar, the time has come to bring them together.

Expand to Cover the Major Forms of Terrestrial Transport

Today, Tesla addresses two relatively small segments of premium sedans and SUVs. With the Model 3, a future compact SUV and a new kind of pickup truck, we plan to address most of the consumer market. A lower cost vehicle than the Model 3 is unlikely to be necessary, because of the third part of the plan described below.

What really matters to accelerate a sustainable future is being able to scale up production volume as quickly as possible. That is why Tesla engineering has transitioned to focus heavily on designing the machine that makes the machine -- turning the factory *itself* into a product. A first principles physics analysis of automotive production suggests that somewhere between a 5 to 10 fold improvement is achievable by version 3 on a roughly 2 year iteration cycle. The first Model 3 factory machine should be thought of as version 0.5, with version 1.0 probably in 2018.

In addition to consumer vehicles, there are two other types of electric vehicle needed: heavy-duty trucks and high passenger-density urban transport. Both are in the early stages of development at Tesla and should be ready for unveiling next year. We believe the Tesla Semi will deliver a substantial reduction in the cost of cargo transport, while increasing safety and making it really fun to operate.

With the advent of autonomy, it will probably make sense to shrink the size of buses and transition the role of bus driver to that of fleet manager. Traffic congestion would improve due to increased passenger areal density by eliminating the center aisle and putting seats where there are currently entryways, and matching acceleration and braking to other vehicles, thus avoiding the inertial impedance to smooth traffic flow of traditional heavy buses. It would also take people all the way to their destination. Fixed summon buttons at existing bus stops would serve those who don't have a phone. Design accommodates wheelchairs, strollers and bikes.

Autonomy

As the technology matures, all Tesla vehicles will have the hardware necessary to be fully self-driving with fail-operational capability, meaning that any given system in the car could break and your car will still drive itself safely. It is important to emphasize that refinement and validation of the software will take much longer than putting in place the cameras, radar, sonar and computing hardware.

Even once the software is highly refined and far better than the average human driver, there will still be a significant time gap, varying widely by jurisdiction, before true self-driving is approved by regulators. We expect that worldwide regulatory approval will require something on the order of 6 billion miles (10 billion km). Current fleet learning is happening at just over 3 million miles (5 million km) per day.

I should add a note here to explain why Tesla is deploying partial autonomy now, rather than waiting until some point in the future. The most important reason is that, when used correctly, it is already significantly safer than a person driving by themselves and it would therefore be morally reprehensible to delay release simply for fear of bad press or some mercantile calculation of legal liability.

According to the recently released 2015 NHTSA report, automotive fatalities increased by 8% to one death every 89 million miles. Autopilot miles will soon exceed twice that number and the system gets better every day. It would no more make sense to disable Tesla's Autopilot, as some have called for, than it would to disable autopilot in aircraft, after which our system is named.

It is also important to explain why we refer to Autopilot as "beta". This is not beta software in any normal sense of the word. Every release goes through extensive internal validation before it reaches any customers. It is called beta in order to decrease complacency and indicate that it will continue to improve (Autopilot is always off by default). Once we get to the point where Autopilot is approximately 10 times safer than the US vehicle average, the beta label will be removed.

Sharing

When true self-driving is approved by regulators, it will mean that you will be able to summon your Tesla from pretty much anywhere. Once it picks you up, you will be able to sleep, read or do anything else enroute to your destination.

You will also be able to add your car to the Tesla shared fleet just by tapping a button on the Tesla phone app and have it generate income for you while you're at work or on vacation, significantly offsetting and at times potentially exceeding the monthly loan or lease cost. This dramatically lowers the true cost of ownership to the point where almost anyone could own a Tesla. Since most cars are only in use by their owner for 5% to 10% of the day, the fundamental economic utility of a true self-driving car is likely to be several times that of a car which is not.

In cities where demand exceeds the supply of customer-owned cars, Tesla will operate its own fleet, ensuring you can always hail a ride from us no matter where you are.

So, in short, Master Plan, Part Deux is:

Create stunning solar roofs with seamlessly integrated battery storage

Expand the electric vehicle product line to address all major segments

Develop a self-driving capability that is 10X safer than manual via massive fleet learning

Enable your car to make money for you when you aren't using it

Master Plan Part 3

The Tesla Team, April 5, 2023



Today, we are publishing Master Plan Part 3, which outlines a proposed path to reach a sustainable global energy economy through end-use electrification and sustainable electricity generation and storage. This paper outlines the assumptions, sources and calculations behind that proposal. Input and conversation are welcome.

How Master Plan 3 works:

240_{TWh}
Storage

30_{TW}
Renewable Power

\$10T
Manufacturing Investment

1/2
The Energy Required

0.21%
Land Area Required

10%
2022 World GDP

ZERO
Insurmountable Resource Challenges

Exhibit 2: Elon Musk Economist Tweet



Exhibit 3: Elon Musk "Funding Secured" Tweet



Exhibit 4: Twitter vs. Elon Musk

IN THE COURT OF CHANCERY OF THE STATE OF
DELAWARE

TWITTER, INC.,

Plaintiff, v.

ELON R. MUSK, X HOLDINGS I, INC., and X HOLDINGS
II, INC.,

Defendants.

Verified Complaint

Plaintiff Twitter, Inc. (“Twitter”), by and through its undersigned counsel, as and for its complaint against defendants Elon R. Musk, X Holdings I, Inc. (“Parent”), and X Holdings II, Inc. (“Acquisition Sub”), alleges as follows:

Nature of the Action

1. In April 2022, Elon Musk entered into a binding merger agreement with Twitter, promising to use his best efforts to get the deal done. Now, less than three months later, Musk refuses to honor his obligations to Twitter and its stockholders because the deal he signed no longer serves his personal interests. Having mounted a public spectacle to put Twitter in play, and having proposed and then signed a seller-friendly merger agreement, Musk apparently believes that he—unlike every other party subject to Delaware contract law—is free to change his mind, trash the company, disrupt its operations, destroy stockholder value, and walk away. This repudiation follows a long list of

material contractual breaches by Musk that have cast a pall over Twitter and its business. Twitter brings this action to enjoin Musk from further breaches, to compel Musk to fulfill his legal obligations, and to compel consummation of the merger upon satisfaction of the few outstanding conditions.

2. Musk, the Chief Executive Officer of Tesla, Inc. and leader of SpaceX and other entities, opened a Twitter account in 2009. His presence on the Twitter platform is ubiquitous. With over 100 million followers, Musk's account is one of the most followed on Twitter, and he has Tweeted more than 18,000 times. He has also suggested he would consider starting his own company to compete with Twitter.

3. On April 25, 2022, Musk, acting through and with his solely-owned entities, Parent and Acquisition Sub, agreed to buy Twitter for \$54.20 per share in cash, for a total of about \$44 billion.

4. That price, presented by Musk on a take-it-or-leave-it basis in an unsolicited public offer, represented a 38% premium over Twitter's unaffected share price. The other terms Musk offered and agreed to were, as he touted, "seller friendly." There is no financing contingency and no diligence condition. The deal is backed by airtight debt and equity commitments. Musk has personally committed \$33.5 billion.

5. After the merger agreement was signed, the market fell. As the Wall Street Journal reported recently, the value of Musk's stake in Tesla, the anchor of his personal wealth,

has declined by more than \$100 billion from its November 2021 peak.

6. So Musk wants out. Rather than bear the cost of the market downturn, as the merger agreement requires, Musk wants to shift it to Twitter's stockholders. This is in keeping with the tactics Musk has deployed against Twitter and its stockholders since earlier this year, when he started amassing an undisclosed stake in the company and continued to grow his position without required notification. It tracks the disdain he has shown for the company that one would have expected Musk, as its would-be steward, to protect. Since signing the merger agreement, Musk has repeatedly disparaged Twitter and the deal, creating business risk for Twitter and downward pressure on its share price.

7. Musk's exit strategy is a model of hypocrisy. One of the chief reasons Musk cited on March 31, 2022 for wanting to buy Twitter was to rid it of the "[c]rypto spam" he viewed as a "major blight on the user experience." Musk said he needed to take the company private because, according to him, purging spam would otherwise be commercially impractical. In his press release announcing the deal on April 25, 2022, Musk raised a clarion call to "defeat[] the spam bots." But when the market declined and the fixed-price deal became less attractive, Musk shifted his narrative, suddenly demanding "verification" that spam was not a serious problem on Twitter's platform, and claiming a burning need to conduct "diligence" he had expressly forsworn.

8. Musk's strategy is also a model of bad faith. While pretending to exercise the narrow right he has under the merger agreement to information for "consummation of the transaction," Musk has been working furiously—albeit fruitlessly—to try to show that the company he promised to buy and not disparage has made material misrepresentations about its business to regulators and investors. He has also asserted, falsely, that consummation of the merger depends on the results of his fishing expedition and his ability to secure debt financing.

9. On July 8, 2022, a little over a month after first using bad-faith pursuit of spam-related evidence to assert a baseless claim of breach, Musk gave Twitter notice purporting to terminate the merger agreement. The notice alleges three grounds for termination: (i) purported breach of information-sharing and cooperation covenants; (ii) supposed "materially inaccurate representations" in the merger agreement that allegedly are "reasonably likely to result in" a Company Material Adverse Effect; and (iii) purported failure to comply with the ordinary course covenant by terminating certain employees, slowing hiring, and failing to retain key personnel.

10. These claims are pretexts and lack any merit. Twitter has abided by its covenants, and no Company Material Adverse Effect has occurred or is reasonably likely to occur. Musk, by contrast, has been acting against this deal since the market started turning, and has breached the merger agreement repeatedly in the process. He has purported to put the deal on "hold" pending satisfaction of imaginary

conditions, breached his financing efforts obligations in the process, violated his obligations to treat requests for consent reasonably and to provide information about financing status, violated his non-disparagement obligation, misused confidential information, and otherwise failed to employ required efforts to consummate the acquisition.

11. Twitter is entitled to specific performance of defendants' obligations under the merger agreement and to secure for Twitter stockholders the benefit of Musk's bargain. Musk and his entities should be enjoined from further breaches, ordered to comply with their obligations to work toward satisfying the few closing conditions, and ordered to close upon satisfaction of those conditions.

The Parties

12. Plaintiff Twitter, Inc. is a Delaware corporation headquartered in San Francisco, California that owns and operates a global platform for real-time self-expression and conversation.

13. Defendant Elon R. Musk is a sophisticated entrepreneur who owns approximately 9.6% of Twitter's stock. He is the CEO of Tesla and leads SpaceX, among other entities he founded or co-founded. Musk is referred to in the merger agreement as Equity Investor, and is the President, Secretary, Treasurer, and sole shareholder of both Parent and Acquisition Sub. Musk signed the merger agreement on behalf of both Parent and Acquisition Sub, and agreed to be a party to the agreement in his individual capacity as

Equity Investor with respect to several “Specified Provisions.” Ex. 1, Preamble.

14. Defendant X Holdings I, Inc., or Parent, is a Delaware corporation formed on April 19, 2022 solely for the purpose of engaging in, and arranging financing for, the transactions contemplated by the merger agreement.

15. Defendant X Holdings II, Inc., or Acquisition Sub, is a Delaware corporation and a wholly owned subsidiary of Parent. Acquisition Sub was formed on April 19, 2022 solely for the purpose of engaging in, and arranging financing for, the transactions contemplated by the merger agreement.

Jurisdiction

16. This Court has subject matter jurisdiction under 10 Del. C. § 341, 8 Del. C. § 111(a), and 6 Del. C. § 2708.

17. Personal jurisdiction over Parent and Acquisition Sub is proper because both are incorporated under the laws of Delaware and consented to jurisdiction by agreeing to “expressly and irrevocably submit[] to the exclusive personal jurisdiction of the Delaware Court of Chancery . . . in the event any dispute arises out of [the merger agreement] or the transactions contemplated by [the merger agreement].” Ex. 1 § 9.10(a).

18. Personal jurisdiction over Musk is proper pursuant to 10 Del. C. § 3104(c)(1) because, among other things, (a) Musk formed Parent and Acquisition Sub, both Delaware corporations wholly owned by Musk, for the sole purpose of acquiring Twitter, a Delaware corporation; and (b) Musk

agreed to undertake “reasonable best efforts” to consummate the contemplated transaction, including by causing Parent and Acquisition Sub to deliver a Certificate of Merger to the Delaware Secretary of State. *Id.* §§ 2.3(a), 6.3(a).

Factual Allegations

I. Musk sets his sights on Twitter

19. Musk is active on Twitter’s platform and has expressed a keen interest in the use and inherent potential of the platform. Starting in January 2022, Musk began purchasing Twitter stock. By March 14, 2022, he had secretly accumulated a substantial position—about 5% of the company’s outstanding shares. SEC regulations required that he disclose that position no later than March 24, 2022. Musk failed to disclose, and instead kept amassing Twitter stock with the market none the wiser. By April 1, 2022, Musk had accumulated about 9.1% of the company’s outstanding shares, still in secret. Not until April 4, 2022 did Musk finally disclose his holdings, which made him Twitter’s largest stockholder. Twitter’s stock price jumped 27% upon the disclosure. Between March 24 and April 4, over 112 million Twitter shares traded in ignorance of Musk’s mounting ownership.

20. Meanwhile, on March 26, 2022, Musk spoke with two Twitter directors, Jack Dorsey and Egon Durban, about the future of social media and the prospect of Musk’s joining the Twitter board. Soon after, Musk told Twitter CEO Parag Agrawal and Twitter board chair Bret Taylor that he had in

mind three options relative to Twitter: join its board, take the company private, or start a competitor.

21. Musk would repeat this statement over the coming days. His contemplation of building a rival platform to Twitter was not a secret. On March 26, 2022, he had Tweeted that he was giving “serious thought” to the idea.

22. In early April, after further discussion among Musk, Agrawal, Taylor, and Twitter director Martha Lane Fox, chair of Twitter’s nominating and corporate governance committee, the Twitter board evaluated Musk’s candidacy as a director. Having considered, among other things, Musk’s interest in the platform, his technical expertise, and the perspectives he could bring, the board offered Musk the position on April 3. Musk accepted, the parties signed a letter agreement, and the agreement was announced on April 5.

23. Not a week later, Musk abruptly changed tack. On April 9, the day his appointment to the board was to become effective, Musk told Twitter he would not join the board. Instead, he would offer to buy the company.

II. Musk offers to buy Twitter

24. On April 13—four days after reversing course on the board seat—Musk texted Taylor that he planned to make an offer to acquire all of Twitter. His unsolicited offer, conveyed by letter later that day, was accompanied by a threat:

I am offering to buy 100% of Twitter for \$54.20 per share in cash, a 54% premium over the day before I began investing

in Twitter and a 38% premium over the day before my investment was publicly announced. My offer is my best and final offer and if it is not accepted, I would need to reconsider my position as a shareholder.

25. The following day, on April 14, Musk announced his offer publicly and noted that it was conditioned on customary business due diligence and financing. At a public event the same day, Musk—whose enormous personal wealth exceeds the capital of most public companies—boasted that he could “technically afford” to purchase Twitter outright.

26. Also on April 14, the Twitter board met to discuss Musk’s proposal. It established a transactions committee composed of independent directors Taylor, Lane Fox, and Patrick Pichette to evaluate the proposal, oversee negotiations, and explore strategic alternatives. The board was assisted in its review by Goldman Sachs and J.P. Morgan as financial advisors, and Simpson Thacher as independent counsel.

27. Faced with Musk’s rapid accumulation of Twitter stock and take-it-or-leave-it offer, and concerned that he might launch a hostile tender offer without notice, the board adopted a customary shareholder rights plan to protect its stockholders from “coercive or otherwise unfair takeover tactics.” The board took this action to reduce the likelihood of a takeover without payment of an appropriate control premium and to ensure that the board had sufficient time to make an informed judgment on Musk’s or any other offer. Under the rights plan’s terms, a single investor or group’s

acquisition of more than 15% of the company's outstanding common stock without board approval gives other stockholders the opportunity to acquire additional stock at a considerable discount. The plan was adopted and announced on April 15, 2022.

28. The board's concerns proved well-grounded. Musk began making all-too-obvious public references to a hostile tender offer:



29. At the same time, Musk worked to strengthen the offer he had made and might make by tender. By April 20, he had personally committed \$21 billion in equity financing and lined up \$25.5 billion of committed debt financing, with \$12.5 billion of that secured by his Tesla stock.

30. Having obtained these commitments, Musk announced in an April 21, 2022 securities filing that his offer was no longer conditioned on financing or subject to due diligence:

At the time of delivery, the Proposal was also subject to the completion of financing and business due diligence, but it is no longer subject to financing as a result of the Reporting Person's receipt of the financing commitments. . . and is no longer subject to business due diligence.

Musk proclaimed himself prepared to begin negotiations “immediately,” and confirmed he was “exploring whether to commence a tender offer.”

31. On Saturday, April 23, 2022, Musk asked to speak with Twitter representatives about his offer. At the direction of the transactions committee, Taylor engaged with Musk, who reiterated that his offer was “best and final” and threatened once again to take it to Twitter’s stockholders directly if the board did not engage immediately.

32. The following day, on Sunday, April 24, 2022, Musk tried again to force Twitter’s hand. He delivered a letter to the board repeating that his \$54.20 per share offer was “best and final,” threatening once more to sell all of his shares if his bid were rejected, and saying he would propose a “seller friendly” merger agreement to be signed before the market opened the next day. Musk’s counsel sent over a draft agreement, reiterated that Musk’s offer was not contingent on any due diligence, and underscored that the form of the proposed agreement was “intended to make this easy on all to get to a deal asap.”

33. The agreement was negotiated through the night and, in the process, became even more seller-friendly. Among the provisions not contained in Musk’s proposal but included at Twitter’s insistence were an undertaking by defendants, including Musk, to “take or cause to be taken . . . all actions and to do, or cause to be done, all things necessary, proper or advisable” to obtain the financing (already committed) to consummate the transaction, Ex. 1 § 6.10(a); a clear disclaimer of any financing condition to

closing, *id.* § 6.10(f); and a right on Twitter's part to request and promptly receive updates from Musk about his progress in obtaining financing, *id.* § 6.10(d). These provisions ensured that financing would be no obstacle to closing and that the company would have the right to stay informed of Musk's progress in arranging his financing.

34. Twitter also negotiated for itself a right to hire and fire employees at all levels, including executives, without having to seek Musk's consent. Musk's initial draft of the merger agreement would have deemed the hiring and firing of an employee at the level of vice president or above a presumptive violation of the ordinary course covenant absent Musk's consent. Twitter successfully struck that provision before signing.

35. Twitter further negotiated to narrow the circumstances under which defendants could escape the deal by claiming a "Company Material Adverse Effect." In addition to excluding, for example, market-wide and industry-wide effects and circumstances and declines in stock price and financial performance, the final definition excluded matters relating to or resulting from Musk's identity or communications, "performance" of the agreement, and any matter disclosed by Twitter in its SEC filings other than the "Risk Factors" and "Forward-Looking Statements" sections of those disclosures. *Id.* Art. I.

36. Finally, and critically, Twitter negotiated for itself a robust right to demand specific performance of the agreement's terms that encompassed the right to compel defendants to close the deal, and ensured that Musk

personally was bound by that provision (among others). *Id.* § 9.9(a)-(b), Preamble.

37. At a board meeting on April 25, 2022, Goldman Sachs and J.P. Morgan each presented their fairness opinions, and the board discussed the agreement. The board ultimately approved the merger agreement and decided to recommend stockholder approval, both because the price was fair to stockholders and because the merger agreement promised a high level of closing certainty. Twitter had taken Musk's claimed "seller friendly" draft agreement and secured other key concessions to make it even more so. Not only were there no financing or diligence conditions, but Musk had already secured debt commitments that together with his personal equity commitment would suffice to fund the purchase.

38. Twitter had been buffeted by Musk's reversals before. For the benefit of stockholders and employees, the board needed assurance that this agreement would stick. It received that assurance in the terms it was able to negotiate.

III. The final, agreed-upon deal terms

39. The terms of the transaction are governed by the merger agreement executed on April 25, 2022.

40. Under the agreement, at closing, Acquisition Sub will merge into Twitter, and Twitter will continue as a private corporation owned by Musk through his wholly owned shell companies. Twitter stockholders will receive \$54.20 per

share in cash, and the company's common stock will be delisted from the New York Stock Exchange.

A. Closing Conditions

41. The conditions to closing are few. The transaction is subject to a majority vote of Twitter's stockholders and to specified regulatory approvals. *Id.* § 7.1. The deal is also conditioned on the non-occurrence of a Company Material Adverse Effect that is continuing at the time of closing. *Id.* § 7.2(c). The agreement contains various representations by Twitter, including that its SEC filings since January 1, 2022, at the time filed or at the time amended or supplemented, are complete and accurate in all material respects, fairly depict the financial condition of the company in all material respects, and were prepared in accordance with GAAP. *Id.* § 4.6. Any inaccuracy in these representations does not excuse closing unless it rises to the level of a Company Material Adverse Effect. *Id.* § 7.2(b).

42. Company Material Adverse Effect is defined as:

any change, event, effect or circumstance which, individually or in the aggregate, has resulted in or would reasonably be expected to result in a material adverse effect on the business, financial condition or results of operations of the Company and its Subsidiaries, taken as a whole

Id. Art. I. As one would expect with a "seller friendly" merger agreement, the contract identifies numerous changes, events, and circumstances expressly excluded

from the determination of whether a Company Material Adverse Effect has occurred:

[C]hanges, events, effects or circumstances which, directly or indirectly, to the extent they relate to or result from the following shall be excluded from, and not taken into account in, the determination of Company Material Adverse Effect:

(i) any condition, change, effect or circumstance generally affecting any of the industries or markets in which the Company or its Subsidiaries operate;

. . .

(iii) general economic, regulatory or political conditions (or changes therein) or conditions (or changes therein) in the financial, credit or securities markets (including changes in interest or currency exchange rates) in the United States or any other country or region in the world;

. . .

(iv) the negotiation, execution, announcement, performance, consummation or existence of this Agreement or the transactions contemplated by this Agreement, including (A) by reason of the identity of Elon Musk, Parent or any of their Affiliates or their respective financing sources, or any communication by Parent or any of its Affiliates or their respective financing sources, including regarding their plans or intentions with respect to the conduct of the business of the Company or any of its Subsidiaries and (B) any litigation, claim or legal proceeding threatened or initiated against Parent, Acquisition Sub, the Company or any of their respective Affiliates, officers or

directors, in each case, arising out of or relating to the this Agreement or the transactions contemplated by this Agreement, and including the impact of any of the foregoing on any relationships with customers, suppliers, vendors, collaboration partners, employees, unions or regulators;

. . .

(viii) any changes in the market price or trading volume of the Company Common Stock, any failure by the Company or its Subsidiaries to meet internal, analysts' or other earnings estimates or financial projections or forecasts for any period, any changes in credit ratings and any changes in analysts' recommendations or ratings with respect to the Company or any of its Subsidiaries (provided that the facts or occurrences giving rise to or contributing to such changes or failure that are not otherwise excluded from the definition of "Company Material Adverse Effect" may be taken into account in determining whether there has been a Company Material Adverse Effect); and

(ix) any matter disclosed in the Company SEC Documents filed by the Company prior to the date of this Agreement (other than any disclosures set forth under the headings "Risk Factors" or "Forward-Looking Statements"). *Id.*

43. The parties thus agreed that any circumstance affecting the market generally or other social media companies would not excuse defendants from closing. Nor would any circumstance arising from the existence or performance of the agreement, or from any communication by Musk, "including the impact of any of the foregoing" on any of

Twitter's relationships with, among others, customers. Likewise, matters that Twitter disclosed in sections of its SEC filings other than the "Risk Factors" and "Forward-Looking Statements" sections cannot constitute a Company Material Adverse Effect. And Twitter's failure to meet financial projections will not excuse closing unless that failure results from an occurrence independently qualifying as a Company Material Adverse Effect (taking into account all of the express exclusions).

44. The agreement also makes clear that financing is not a condition to closing:

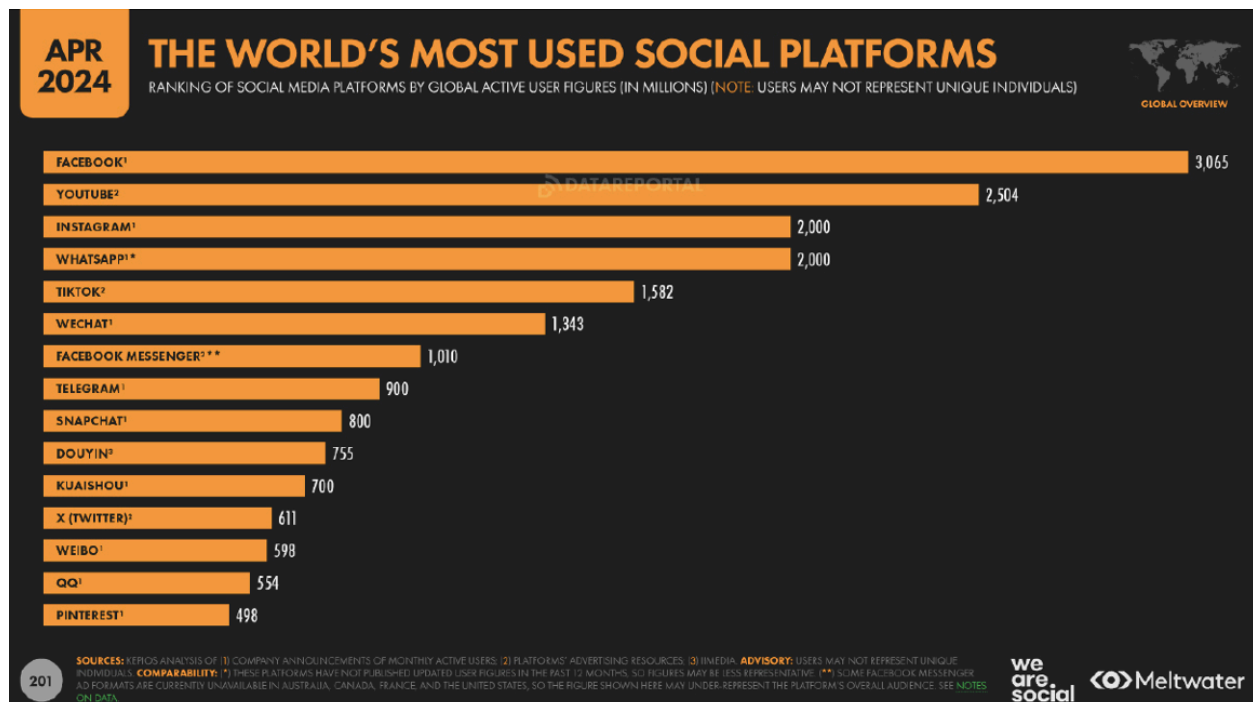
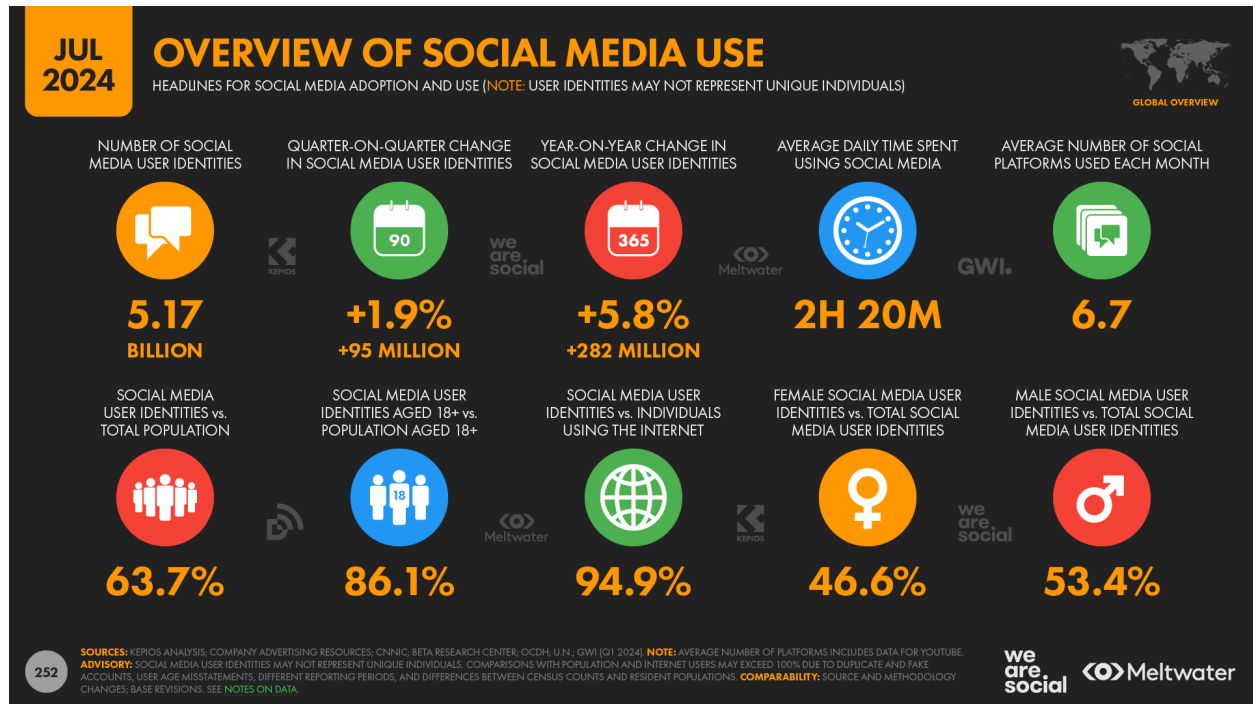
Notwithstanding anything contained in this Agreement to the contrary, the Equity Investor, Parent and Acquisition Sub each acknowledge and affirm that it is not a condition to the Closing or to any of its obligations under this Agreement that the Equity Investor, Parent, Acquisition Sub and/or any of their respective Affiliates obtain any financing (including the Debt Financing) for any of the transactions contemplated by this Agreement.

Id. § 5.4; see also *id.* § 6.10(f).

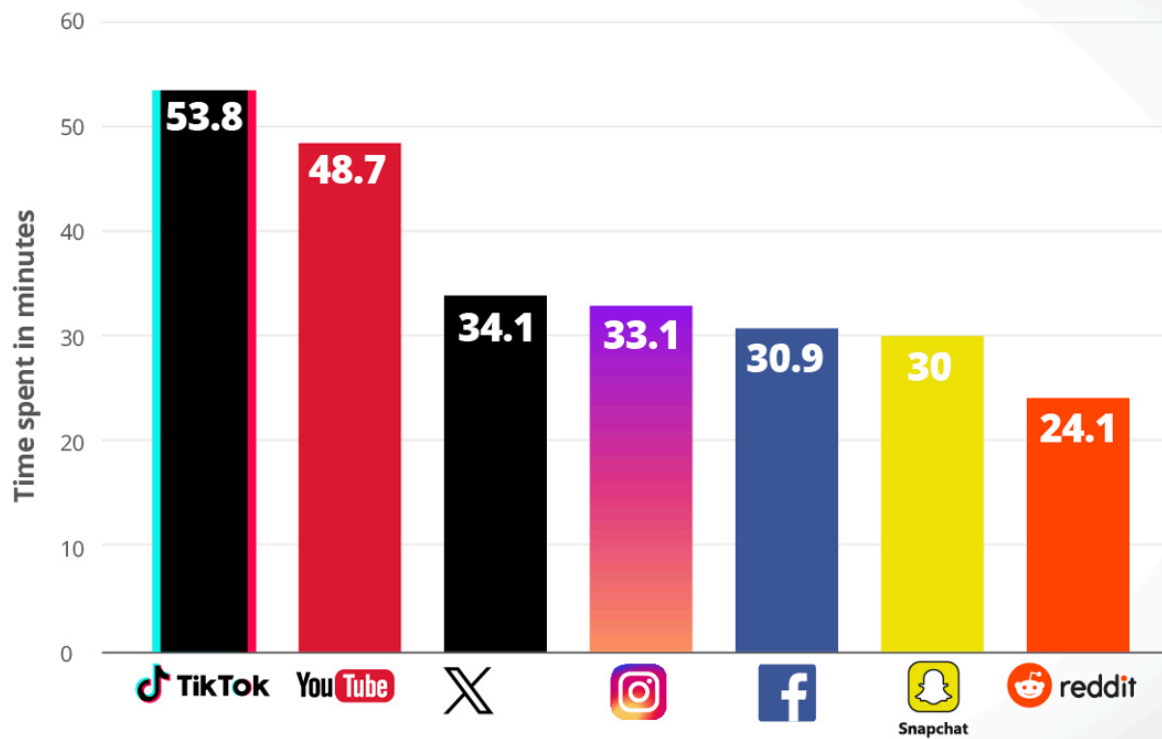
45. Nor is there any diligence condition. Indeed, each of Parent and Acquisition Sub represents that it "conducted, to its satisfaction, its own independent investigation, review and analysis of the business, results of operations, prospects, condition (financial or otherwise) or assets of the Company and its Subsidiaries," and that, in determining to proceed with the merger, each "relied solely on the results of its own independent review and analysis and the

covenants, representations and warranties of the Company” in the merger agreement. *Id.* § 5.11. Parent and Acquisition Sub further acknowledge that “neither the Company nor any of its Subsidiaries, nor any other Person, makes or has made or is making any express or implied representation or warranty with respect to the Company or any of its Subsidiaries or their respective business or operations, in each case, other than those expressly given solely by the Company in Article IV,” and they represent that in agreeing to the merger they were not relying on “any express or implied representation or warranty, or the accuracy or the completeness of the representations and warranties” in the merger agreement about Twitter and its business and its operations “other than those expressly given solely by the Company in Article IV.” *Id.*

Exhibit 5: Social Media Use Statistics



Average time spent per day on select social media platforms in the United States in 2023
(in minutes)

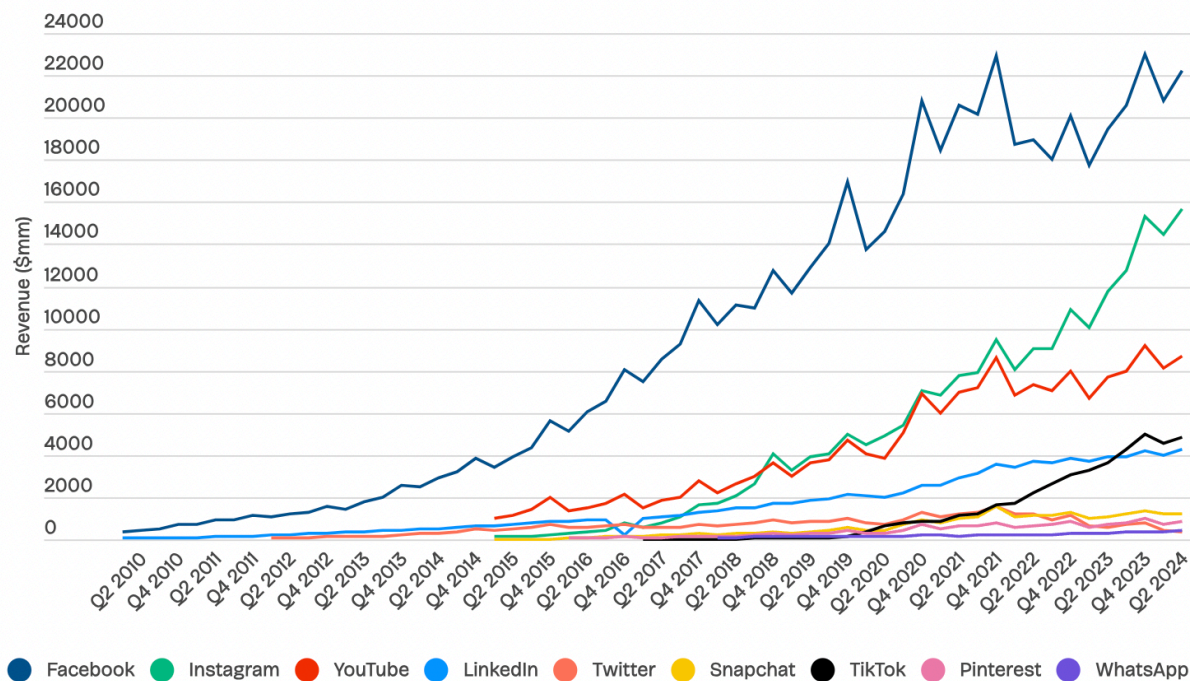


* More than 4 hours a day in total; the typical social media user visits an average of 6.7 social platforms each month

Statista 2024

Exhibit 6: Social Media Platforms Revenues and Users

Social App Revenues 2010 to 2023 (\$mm)



Source: Company data and internal analysis for Instagram and TikTok revenue

DOWNLOAD CHART

Business of Apps

Social Networking Users by App 2023 (bn)

App	Users (bn)
Facebook	3
YouTube	2.7
WhatsApp	2.4
Instagram	2.4
TikTok	1.6
WeChat	1
Reddit	0.85
Snapchat	0.8
QQ	0.57
Pinterest	0.46
Twitter	0.42
LinkedIn	0.31

Exhibit 7: Twitter Financial Statements – As of March 31, 2022

CONSOLIDATED BALANCE SHEETS (In thousands, except par value) (Unaudited)

	March 31, 2022	December 31, 2021
Assets		
Current assets:		
Cash and cash equivalents	\$ 2,283,308	\$ 2,186,549
Short-term investments	3,978,645	4,207,133
Accounts receivable, net of allowance for doubtful accounts of \$13,947 and \$15,278	948,142	1,217,404
Prepaid expenses and other current assets	265,973	266,484
Assets held for sale	—	40,800
Total current assets	7,476,068	7,918,370
Property and equipment, net	2,150,581	2,082,160
Operating lease right-of-use assets	1,272,435	1,195,124
Intangible assets, net	59,725	69,324
Goodwill	1,298,462	1,301,520
Deferred tax assets, net	941,883	1,148,573
Other assets	351,803	344,445
Total assets	\$ 13,550,957	\$ 14,059,516
Liabilities and stockholders' equity		
Current liabilities:		
Accounts payable	\$ 189,524	\$ 203,171
Accrued and other current liabilities	738,778	918,350
Operating lease liabilities, short-term	208,405	222,346
Total current liabilities	1,136,707	1,343,867
Convertible notes, long-term	3,561,067	3,559,023
Senior notes, long-term	1,683,114	693,996
Operating lease liabilities, long-term	1,173,879	1,071,209
Deferred and other long-term tax liabilities, net	41,227	40,691
Other long-term liabilities	50,047	43,531
Total liabilities	7,646,041	6,752,317
Commitments and contingencies (Note 15)		
Stockholders' equity:		
Preferred stock, \$0.000005 par value—200,000 shares authorized; none issued and outstanding	—	—
Common stock, \$0.000005 par value—5,000,000 shares authorized; 763,578 and 799,384 shares issued and outstanding	4	4
Additional paid-in capital	6,677,294	8,432,112
Treasury stock, at cost—0 and 120 shares	—	(5,295)
Accumulated other comprehensive loss	(155,522)	(117,320)
Accumulated deficit	(616,860)	(1,002,302)
Total stockholders' equity	5,904,916	7,307,199
Total liabilities and stockholders' equity	\$ 13,550,957	\$ 14,059,516

TWITTER, INC. CONSOLIDATED STATEMENTS OF INCOME (In thousands, except per share data) (Unaudited)

	Three Months Ended March 31,	
	2022	2021
Revenue	\$ 1,200,984	\$ 1,036,018
Costs and expenses		
Cost of revenue	507,450	381,008
Research and development	371,695	250,709
Sales and marketing	299,809	234,592
General and administrative	149,863	117,527
Total costs and expenses	1,328,817	983,836
Income (loss) from operations	(127,833)	52,182
Interest expense	(15,444)	(13,185)
Interest income	7,862	11,001
Other income (expense), net	(6,506)	6
Gain on sale of asset group	970,474	—
Income before income taxes	828,653	50,004
Provision (benefit) for income taxes	315,367	(18,001)
Net income	\$ 513,286	\$ 68,005
Net income per share:		
Basic	\$ 0.66	\$ 0.09
Diluted	\$ 0.61	\$ 0.08
Numerator used to compute net income per share:		
Basic	\$ 513,286	\$ 68,005
Diluted	\$ 515,313	\$ 68,005
Weighted-average shares used to compute net income per share:		
Basic	778,937	795,633
Diluted	838,590	872,187

TWITTER, INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(In thousands)
(Unaudited)

	Three Months Ended March 31,			
	2022		2021	
	Shares	Amount	Shares	Amount
Common stock				
Balance, beginning of period	799,384	\$ 4	796,000	\$ 4
Issuance of common stock in connection with RSU vesting	4,121	—	4,400	—
Exercise of stock options	5	—	523	—
Shares withheld related to net share settlement of equity awards	(114)	—	(155)	—
Repurchases of common stock	(39,788)	—	(2,723)	—
Other activities	(30)	—	38	—
Balance, end of period	763,578	\$ 4	798,083	\$ 4
Additional paid-in capital				
Balance, beginning of period	—	\$ 8,432,112	—	\$ 9,167,138
Exercise of stock options	—	77	—	1,957
Shares withheld related to net share settlement of equity awards	—	(4,190)	—	(10,569)
Stock-based compensation	—	199,210	—	125,399
Purchase of convertible note hedge	—	—	—	(213,469)
Tax related to purchase of convertible note hedge	—	—	—	49,252
Issuance of warrants	—	—	—	161,144
Repurchases of common stock	—	(1,349,915)	—	(161,552)
Purchase of forward contract under accelerated share repurchase agreements	—	(600,000)	—	—
Cumulative-effect adjustment from adoption of new accounting standard	—	—	—	(567,547)
Balance, end of period	—	\$ 6,677,294	—	\$ 8,551,763
Treasury stock				
Balance, beginning of period	—	\$ (5,295)	—	\$ (5,297)
Retirement of treasury stock	—	5,295	—	5,297
Repurchases of common stock	—	—	—	(5,297)
Balance, end of period	—	\$ —	—	\$ (5,297)
Accumulated other comprehensive loss				
Balance, beginning of period	—	\$ (117,320)	—	\$ (66,094)
Other comprehensive loss	—	(38,202)	—	(30,838)
Balance, end of period	—	\$ (155,522)	—	\$ (96,932)
Accumulated deficit				
Balance, beginning of period	—	\$ (1,002,302)	—	\$ (1,125,669)
Repurchases of common stock	—	(127,844)	—	—
Cumulative-effect adjustment from adoption of new accounting standard	—	—	—	344,776
Net income	—	513,286	—	68,005
Balance, end of period	—	\$ (616,860)	—	\$ (712,888)
Total stockholders' equity	763,578	\$ 5,904,916	798,083	\$ 7,736,650

End Notes

Aiello, C. (2024). *How Elon Musk Leaned on Starlink to Achieve Profitability at SpaceX*. [online] Inc. Available at: <https://www.inc.com/chloe-aiello/how-elon-musk-leaned-on-starlink-to-achieve-profitability-at-spacex.html>.

Alamalhodaiei, A. (2024). *Exclusive: Internal pre-Starlink SpaceX financials show big spending on moonshot bets*. [online] TechCrunch. Available at: <https://techcrunch.com/2024/04/11/internal-pre-starlink-spacex-financials-show-big-spending-on-moonshot-bets/>.

Alba, D., Love, J., Kao, J. and Nicoletti, L. (2024). *Elon Musk Is Now X's Biggest Promoter of Anti-Immigrant Conspiracies*. [online] Bloomberg. Available at:
https://www.bloomberg.com/graphics/2024-musk-x-election-influence-immigration/?accessToken=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzb3VyY2UiOiJTdWJzY3JpYmVrR2lmdGVkQXJ0aWNsZSIsImhhbmhhdCI6MTcyOTc3NmM1OCwiZXhwIjoxNzMwMzg5MTU0LCJpcnRpY2xiSWQiOiJTTTFVYOFhEV1JHRzAwMCIsImjb25uZWNO0SWQiOiJBQkE4QTQ2RTQ5MzE0RUVBQjcwM0NDQzU0MkQ4ODE1MSJ9.hhXgVuBLH87875Jl2EO10Xq0EbdGr2_YmjxHMBJ2UbK.

Albergotti, R. (2023). *The secret history of Elon Musk, Sam Altman, and OpenAI* | Semafor. [online] www.semafor.com. Available at: <https://www.semafor.com/article/03/24/2023/the-secret-history-of-elon-musk-sam-altman-and-openai>.

Allen, H. (2024). *Donald Trump Elon Musk X Interview | Rev.* [online] Rev. Available at: <https://www.rev.com/blog/transcripts/elon-musk-and-donald-trump-interview>.

AP NEWS. (2022). *Timeline of billionaire Elon Musk's bid to control Twitter*. [online] Available at: <https://apnews.com/article/twitter-elon-musk-timeline-c6b09620ee0905e59df9325ed042a609>.

AP News. (2023). *Tesla moves forward with a plan to build an energy-storage battery factory in China*. [online] Available at: <https://apnews.com/article/china-tesla-megapack-battery-factory-shanghai-306b4f6678c6827080a7bf49d55d12ee>.

Ap.org. (2024). AP. [online] Available at: <https://newsroom.ap.org/editorial-photos-videos/detail?itemid=21ef8baf66e5da11af9f0014c2589dfb&mediatype=photo> [Accessed 28 Oct. 2024].

Apple.news. (2024). *Elon Musk Has Turned X Into His Personal Political Playground – WIRED*. [online] Available at: <https://apple.news/AJE27qKtzTFag01JYUhsjSg> [Accessed 28 Oct. 2024].

Ashraf, A. (n.d.). *Tesla CEO Elon Musk Slams Environmental Activists Protesting Giga Berlin Expansion: 'They're Dumber Than A Doorstop Or...Puppets'* - Tesla (NASDAQ:TSLA). [online] Benzinga. Available at:

<https://www.benzinga.com/news/24/03/37641764/tesla-ceo-elon-musk-slams-environmental-activists-protesting-giga-berlin-expansion-theyre-dumber-tha>.

Bansal, A. (2023). *The Future of Cars: Safer, Smarter and More Connected*. [online] IEEE-USA InSight. Available at: <https://insight.ieeeusa.org/articles/the-future-of-cars-safer-smarter-and-more-connected/>.

Bauer, P. (2024). *Britannica Money*. [online] www.britannica.com. Available at: <https://www.britannica.com/money/Zip2>.

Bekalu, M.A., McCloud, R.F. and Viswanath, K. (2019). Association of Social Media Use with Social Well-Being, Positive Mental Health, and Self-Rated Health: Disentangling Routine Use from Emotional Connection to Use. *Health Education & Behavior*, [online] 46(2), pp.69-80. doi:<https://doi.org/10.1177/1090198119863768>.

Belanger, A. (2024). *Court: Section 230 doesn't shield TikTok from Blackout Challenge death suit*. [online] Ars Technica. Available at: <https://arstechnica.com/tech-policy/2024/08/court-section-230-doesnt-shield-tiktok-from-blackout-challenge-death-suit/> [Accessed 28 Oct. 2024].

BIOGRAPHY (2022). *Elon Musk - Tesla, Age & Family*. [online] Biography. Available at: <https://www.biography.com/business-leaders/elon-musk>.

Bruner, R. (2022). *A Complete Timeline of Elon Musk's Business Endeavors*. [online] Time. Available at: <https://time.com/6170834/elon-musk-business-timeline-twitter/>.

Bursztynsky, L.K. (2022). *Elon Musk wins shareholder lawsuit over Tesla's \$2.6 billion SolarCity acquisition*. [online] CNBC. Available at: <https://www.cnbc.com/2022/04/27/elon-musk-wins-shareholder-lawsuit-over-the-companys-2point6-billion-solarcity-acquisition.html>.

Business, J.W., CNN (n.d.). *SpaceX gets \$1.4 billion contract for five more NASA astronaut launches*. [online] CNN. Available at: <https://www.cnn.com/2022/08/31/tech/nasa-spacex-crew-launches-contract-extension-scn/index.html>.

Carpenter, S. (2024). *NHTSA is investigating 22 incidents with Waymo self-driving cars*. [online] Spectrumnews1.com. Available at: <https://spectrumnews1.com/ca/southern-california/transportation/2024/05/14/nhtsa-is-investigating-22-incidents-with-waymo-self-driving-cars> [Accessed 28 Oct. 2024].

Casiano, L. (2020). *Elon Musk reopens Tesla factory closed by coronavirus: 'If anyone is arrested, I ask that it only be me'*. [online] Fox News. Available at: <https://www.foxnews.com/auto/tesla-elon-musk-factory-reopens-california-coronavirus-tweet> [Accessed 28 Oct. 2024].

Chang, K. (2021). SpaceX Wins NASA \$2.9 Billion Contract to Build Moon Lander. *The New York Times*. [online] 16 Apr. Available at: <https://www.nytimes.com/2021/04/16/science/spacex-moon-nasa.html?smid=nytcore-ios-share&referringSource=articleShare&sgrp=c-cb> [Accessed 28 Oct. 2024].

Cnn.com. (2024). Available at: <https://www.cnn.com/interactive/2017/02/specials/elon-musk-timeline/elon-musk-timeline.html>.

College of Humanities and Social Sciences. (2024). *What are the Risks and Rewards of Autonomous Vehicles?* [online] Available at: <https://chass.ncsu.edu/news/2024/03/28/what-are-the-risks-and-rewards-of-autonomous-vehicles/>.

Companies Market Cap (2024). *Tesla (TSLA) - Market capitalization*. [online] companiesmarketcap.com. Available at: <https://companiesmarketcap.com/tesla/marketcap/>.

Congressional Research Service (2021). *Section 230: An Overview*. [online] Available at: <https://crsreports.congress.gov/product/pdf/R/R46751>.

Counts, A. and Nakano, E. (2023). *Twitter's Surge in Harmful Content Keeps Advertiser Away*. [online] Time. Available at: <https://time.com/6295711/twitters-hate-content-advertisers/>.

Crunchbase. (2023). *Crunchbase*. [online] Available at: <https://www.crunchbase.com/organization/solarcity> [Accessed 28 Oct. 2024].

Curry, D. (2022). *Social Networking App Revenue and Usage Statistics (2022)*. [online] Business of Apps. Available at: <https://www.businessofapps.com/data/social-app-market/>.

Daily News. (2008). *SpaceX awarded \$1.6 billion contract*. [online] Available at: <https://www.dailynews.com/2008/12/24/spacex-awarded-16-billion-contract/>.

Danga, B. (2024). *Elon Musk Heralds DOGE as 'the People's Crypto' at Giga Berlin* Read more on DailyCoin: <https://dailycoin.com/elon-musk-heralds-doge-as-the-peoples-crypto-at-giga->

berlin/. [online] DailyCoin. Available at: <https://dailycoin.com/elon-musk-heralds-doge-as-the-peoples-crypto-at-giga-berlin/>.

DataReportal (2024). *Global Social Media Stats*. [online] DataReportal – Global Digital Insights. Available at: <https://datareportal.com/social-media-users>.

Digital Watch Observatory. (2024). *Elon Musk pushes for AI safety law in California | Digital Watch Observatory*. [online] Available at: <https://dig.watch/updates/elon-musk-pushes-for-ai-safety-law-in-california> [Accessed 28 Oct. 2024].

Duboust, O. (2024). *Elon Musk predicts human intelligence 'will be 1%' when AI takes over*. [online] euronews. Available at: <https://www.euronews.com/next/2024/05/10/elon-musk-predicts-ai-will-overtake-humans-to-the-point-that-biological-intelligence-will->

EcoWatch. (n.d.). *SolarCity 2023 Review (History, Lawsuits & Tesla Acquisition)*. [online] Available at: <https://www.ecowatch.com/solar/reviews/solarcity>.

Editor BizNews (2024). 🗣️ *Elon Musk cosies up to Trump in glitch-ridden X livestream*. [online] BizNews.com. Available at: <https://www.biznews.com/global-citizen/2024/08/13/elon-musk-trump-x-livestream> [Accessed 28 Oct. 2024].

Egan, M. (2024). *Elon Musk's X is worth nearly 80% less than when he bought it, Fidelity estimates*. [online] CNN. Available at: https://www.cnn.com/2024/10/02/business/elon-musk-twitter-x-fidelity?cid=ios_app [Accessed 28 Oct. 2024].

Electronic Frontier Foundation (2023). *Section 230 of the Communications Decency Act*. [online] Electronic Frontier Foundation. Available at: <https://www.eff.org/issues/cda230>.

Fcc.gov. (2023). *FCC Reaffirms Decision to Reject Starlink Application for Nearly \$900 Million in Subsidies*. [online] Available at: <https://www.fcc.gov/document/fcc-reaffirms-decision-reject-starlink-application-nearly-900-million-subsidies>.

Fortune. (n.d.). *Elon Musk Says Tesla Vehicles Will Drive Themselves in Two Years*. [online] Available at: <https://fortune.com/2015/12/21/elon-musk-interview/>.

Fortune. (n.d.). *The quality of Twitter conversation has 'decayed' after Elon Musk's takeover, report finds*. [online] Available at: <https://fortune.com/2022/11/08/elon-musk-twitter-conversation-quality-decayed/>.

Fried, N. (2021). *Why Section 230 Isn't Really a Good Samaritan Provision*. [online] Available at:

<https://www.congress.gov/117/meeting/house/111407/documents/HHRG-117-IF16-20210325-SD013.pdf>.

Fung, B. (2023). *These 26 words 'created the internet.' Now the Supreme Court may be coming for them* | CNN Business. [online] CNN. Available at: <https://www.cnn.com/2023/02/18/tech/section-230-explainer/index.html>.

Gelles, D., Stewart, J.B., Silver-Greenberg, J. and Kelly, K. (2018). Elon Musk Details 'Excruciating' Personal Toll of Tesla Turmoil. *The New York Times*. [online] 17 Aug. Available at: <https://www.nytimes.com/2018/08/16/business/elon-musk-interview-tesla.html>.

Gene (2017). *Tesla is developing its own custom AI chip, says Elon Musk*. [online] TESLARATI. Available at: <https://www.teslarati.com/tesla-developing-custom-ai-chip-says-elon-musk/> [Accessed 28 Oct. 2024].

Geuens, R. (2024). *What's the average time spent on social media each day?* (2024). [online] SOAX. Available at: <https://soax.com/research/time-spent-on-social-media>.
GLOBAL TRANSPARENCY REPORT H1 2024. (n.d.). Available at: <https://transparency.x.com/content/dam/transparency-twitter/2024/x-global-transparency-report-h1.pdf>.

Hals, T. and Hals, T. (2023). Court upholds Musk's win in \$13 billion lawsuit over Tesla-SolarCity deal. *Reuters*. [online] 6 Jun. Available at: <https://www.reuters.com/legal/court-upholds-ruling-musk-over-tesla-solarcity-deal-2023-06-06/>.

Hao, K. (2021). *The Facebook whistleblower says its algorithms are dangerous. Here's why*. [online] MIT Technology Review. Available at: <https://www.technologyreview.com/2021/10/05/1036519/facebook-whistleblower-frances-haugen-algorithms/>.

Hawkins, A.J. (2024). *Tesla Cybercab announced: Elon Musk's robotaxi is finally here*. [online] The Verge. Available at: <https://www.theverge.com/2024/10/10/24265530/tesla-robotaxi-elon-musk-features-range-price-release-date>.

Hetzner, C. (2024). *Elon Musk forced to step in to resolve Tesla Cybertruck owner's hellish experience*. [online] Fortune. Available at: <https://fortune.com/2024/07/28/elon-musk-tesla-cybertruck-quality-warranty-recall-lemon/>.

hlr (2023). *Twitter, Inc. v. Taamneh*. [online] Harvard Law Review. Available at: <https://harvardlawreview.org/print/vol-137/twitter-inc-v-taamneh/>.

Horton, J. (2024). *Donald Trump's chat with Elon Musk on X fact-checked*. [online] Bbc.com. Available at: <https://www.bbc.com/news/articles/c0jpn2q76n1o>.

Howarth, J. (2024). *Top 27 Social Media Platforms (May 2023)*. [online] Exploding Topics. Available at: <https://explodingtopics.com/blog/top-social-media-platforms>.

Howell, E. (2022). *SpaceX: First Private Flights to Space Station*. [online] Space.com. Available at: <https://www.space.com/18853-spacex.html>.

Hubbard, H. (2022). Twitter has lost 50 of its top 100 advertisers since Elon Musk took over, report says. *NPR*. [online] 25 Nov. Available at: <https://www.npr.org/2022/11/25/1139180002/twitter-loses-50-top-advertisers-elon-musk>.

Hutchinson, A. (2024). *Data Suggests X Could Post a Significant Loss in 2024*. [online] Social Media Today. Available at: <https://www.socialmediatoday.com/news/x-formerly-twitter-could-post-a-significant-loss-in-2024/730370/>.

Ingram, J. and May, M. (2024). *The X factor: How Trump ally Elon Musk is using social media to prime voter mistrust ahead of 2024 election*. [online] Cbsnews.com. Available at: <https://www.cbsnews.com/news/elon-musk-trump-social-media-election-2024/>.

Jalopnik. (2023). *Celebrate With Us: Elon Has Been Promising Self-Driving Cars For Ten Years*. [online] Available at: <https://jalopnik.com/elon-musk-tesla-self-driving-cars-anniversary-autopilot-1850432357>.

Justia Law. (n.d.). *Gonzalez v. Google LLC, 598 U.S. ____ (2023)*. [online] Available at: <https://supreme.justia.com/cases/federal/us/598/21-1333/>.

Kann, S. and Carusone, A. (2022). *In less than a month, Elon Musk has driven away half of Twitter's top 100 advertisers*. [online] Media Matters for America. Available at: <https://www.mediamatters.org/elon-musk/less-month-elon-musk-has-driven-away-half-twitters-top-100-advertisers>.

Kelly, S.M. (2024). *Elon Musk says AI will take all our jobs | CNN Business*. [online] CNN. Available at: <https://www.cnn.com/2024/05/23/tech/elon-musk-ai-your-job/index.html>.

Khan, N. (n.d.). *Elon Musk Biography*. [online] VEDANTU. Available at: <https://www.vedantu.com/biography/elon-musk>.

Kirschner, T.B., Kylie (n.d.). *Members of the PayPal Mafia include tech titans like Elon Musk, Peter Thiel, and Reid Hoffman. Here's where they are now*. [online] Business Insider. Available at: <https://www.businessinsider.com/paypal-mafia-members-elon-musk>.

peter-thiel-reid-hoffman-companies#luke-nosek-paypal-cofounder-and-vice-president-of-marketing-and-strategy-9.

Krietzberg, I. (2023). *Twitter to X: The Complete Timeline of Elon Musk's Twitter Purchase*. [online] TheStreet. Available at: <https://www.thestreet.com/technology/twitter-to-x-the-complete-timeline-of-elon-musks-twitter-purchase>.

Kutner, A. (n.d.). *Top Five Dangers of Self-Driving Cars | Adam S. Kutner*. [online] Adam S. Kutner, Injury Attorneys. Available at: <https://www.askadamskutner.com/auto-accident/self-driving-cars/>.

Lambert, F. (2015). *Understanding Tesla's self-driving features: The Autopilot*. [online] Electrek. Available at: <https://electrek.co/2015/06/16/understanding-teslas-self-driving-features-the-autopilot/>.

Ludlow, E. and Tan, G. (2023). *SpaceX Eyes \$15 Billion in Sales Next Year on Starlink Strength*. [online] Bloomberg.com. Available at: <https://www.bloomberg.com/news/articles/2023-11-06/spacex-eyes-15-billion-in-sales-next-year-on-starlink-strength?sref=y3YMCJ4e> [Accessed 28 Oct. 2024].

MacCarthy, M. (2024). *The evolving safety and policy challenges of self-driving cars*. [online] Brookings. Available at: <https://www.brookings.edu/articles/the-evolving-safety-and-policy-challenges-of-self-driving-cars/>.

March 2019, E.H. 14 (2020). *SpaceX's Dragon: First Private Spacecraft to Reach Space Station*. [online] Space.com. Available at: <https://www.space.com/18852-spacex-dragon.html>.

Matousek, M. (2018). *Tesla is experiencing a painful year – here's everything that has gone wrong so far*. [online] Business Insider. Available at: <https://www.businessinsider.com/tesla-challenges-in-2018-2018-4>.

Mattar, W. (2022). *Potential Problems with Self-Driving Cars | William Mattar*. [online] William Mattar. Available at: https://williammattar.com/practice-areas/self-driving-car-accident-attorney/potential-problems-with-self-driving-cars/?utm_content=organic_direct.

McFarland, M. (2014). Elon Musk: 'With artificial intelligence we are summoning the demon.' *Washington Post*. [online] 24 Oct. Available at: <https://www.washingtonpost.com/news/innovations/wp/2014/10/24/elon-musk-with-artificial-intelligence-we-are-summoning-the-demon/>.

Means, L. (2022). *NASA Awards SpaceX Second Contract Option for Artemis Moon Landing - NASA*. [online] Nasa. Available at: <https://www.nasa.gov/humans-in-space/nasa-awards-spacex-second-contract-option-for-artemis-moon-landing/>.

Merano, M. (2024). *Tesla aims for 2M+ Cybercab production annually*. [online] TESLARATI. Available at: <https://www.teslarati.com/tesla-cybercab-2-million-production/> [Accessed 28 Oct. 2024].

Metz, C. (2016). *Inside OpenAI, Elon Musk's Wild Plan to Set Artificial Intelligence Free*. [online] Wired. Available at: <https://www.wired.com/2016/04/openai-elon-musk-sam-altman-plan-to-set-artificial-intelligence-free/>.

Metz, C. and Boudette, N.E. (2021). Inside Tesla as Elon Musk Pushed an Unflinching Vision for Self-Driving Cars. *The New York Times*. [online] 6 Dec. Available at: <https://www.nytimes.com/2021/12/06/technology/tesla-autopilot-elon-musk.html>.

Musk, E. (2009). *Elon Musk (@elonmusk) | Twitter*. [online] Twitter.com. Available at: https://twitter.com/elonmusk?ref_src=twsrc%5Egoogle%7Ctwcamp%5Eserp%7Ctwgr%5Eauthor.

Musk, E. (2024). *The Secret Tesla Motors Master Plan (just between you and me) | Tesla*. [online] Tesla. Available at: <https://www.tesla.com/secret-master-plan>.

NASA. (2012). *NASA's Commercial Crew Program Progressing for Future of U.S. Human Spaceflight - NASA*. [online] Available at: <https://www.nasa.gov/general/nasas-commercial-crew-program-progressing-for-future-of-u-s-human-spaceflight/>.

Nasdaq.com. (2024). *NASDAQ Composite Index (COMP) Historical Data | Nasdaq*. [online] Available at: https://www.nasdaq.com/market-activity/index/comp/historical?page=7&rows_per_page=100&timeline=y5 [Accessed 28 Oct. 2024].

NBC News. (n.d.). *Days before the midterms, Twitter lays off employees who fight misinformation*. [online] Available at: <https://www.nbcnews.com/tech/misinformation/twitter-fires-employees-fight-misinformation-midterm-elections-rcna55750>.

New-Vehicle Quality Mainly Dependent on Trouble-Free Technology, J.D. Power Finds Dodge and Kia Rank Highest in a Tie for Overall Initial Quality; Genesis Ranks Highest among Premium Brands; Tesla Profiled for First Time. (n.d.). Available at: <https://www.jdpower.com/sites/default/files/file/2020-06/2020070%20U.S.%20IQS.pdf>.

News, P. (2024). *Musk's X asks to be reinstated in Brazil after complying with judge's orders, source tells AP*. [online] PBS News. Available at: <https://www.pbs.org/newshour/world/musks-x-asks-to-be-reinstated-in-brazil-after-complying-with-judges-orders-source-tells-ap>.

NHTSA (2021). *Automated Vehicles for Safety* | NHTSA. [online] www.nhtsa.gov. Available at: <https://www.nhtsa.gov/vehicle-safety/automated-vehicles-safety>.

Openai.com. (2024). *OpenAI and Elon Musk*. [online] Available at: <https://openai.com/index/openai-elon-musk/>.

Paraguassu, L., Benedito, L.M. and Brito, R. (2024). *Brazil watchdog moves to block access to Elon Musk's X after court order*. [online] Reuters. Available at: <https://www.reuters.com/technology/lula-says-musk-must-respect-brazils-top-court-x-braces-shutdown-2024-08-30/>.

Paredes, R. (2023). *Autonomous Vehicle Safety: The Future of Driving*. [online] SafetyCulture. Available at: <https://safetyculture.com/topics/autonomous-vehicle-safety/>.

PBS NewsHour. (2022). *Elon Musk ordered to abide by SEC settlement over 2018 tweets*. [online] Available at: <https://www.pbs.org/newshour/economy/elon-musk-ordered-to-abide-by-sec-settlement-over-2018-tweets>.

Perrigo, B. (2021). *Inside Frances Haugen's Decision to Take on Facebook*. [online] Time. Available at: <https://time.com/6121931/frances-haugen-facebook-whistleblower-profile/>.

Picker, T.K. (2018). *Elon Musk's tweet about going private costs Tesla short sellers \$1.3 billion*. [online] CNBC. Available at: <https://www.cnbc.com/2018/08/07/elon-musks-tweet-about-going-private-costs-tesla-short-sellers-more-t.html>.

Protected]Om, email, archive →, P. and and Twitter, F.B. on (n.d.). *Inside SpaceX's Lucrative New Government Satellite Program*. [online] The Information. Available at: <https://www.theinformation.com/articles/inside-spacexs-lucrative-new-government-satellite-program>.

Pultarova, T. and Howell, E. (2024). *Starlink: SpaceX's satellite internet project*. [online] Space.com. Available at: <https://www.space.com/spacex-starlink-satellites.html>.
Q.ai - Powering a Personal Wealth Movement (2023). *SpaceX To Raise \$750 Million In Its Latest Round Of Funding*. *Forbes*. [online] 10 Jan. Available at: <https://www.forbes.com/sites/qai/2023/01/10/spacex-to-raise-750-million-in-its-latest-round-of-funding/> [Accessed 28 Oct. 2024].

Ray, S. (n.d.). *SpaceX Reportedly Valued At Around \$210 Billion In Planned Secondary Market Share Sale*. [online] Forbes. Available at: <https://www.forbes.com/sites/siladityaray/2024/06/27/spacex-reportedly-valued-at-around-210-billion-in-planned-secondary-market-share-sale/>.

Reed, E. and Diongson, D. (2020). *History of Tesla: Timeline and Facts*. [online] TheStreet. Available at: <https://www.thestreet.com/technology/history-of-tesla-15088992>.

Reuters (2024). *Brazil's top court imposes new fine before allowing X to resume service*. [online] Reuters. Available at: <https://www.reuters.com/technology/brazils-top-court-rules-x-must-pay-pending-fines-resume-service-2024-09-27/>.

Risen, T. (2014). *Will Tesla's 'D' Be A Self-Driving Car?* [online] US News & World Report. Available at: <https://www.usnews.com/news/articles/2014/10/09/will-teslas-d-be-a-self-driving-car> [Accessed 28 Oct. 2024].

Roof, K. and Johnson, E. (2024). *SpaceX to Sell Insider Shares at \$112 Apiece, Valuing Itself at \$210 Billion*. [online] Bloomberg.com. Available at: <https://www.bloomberg.com/news/articles/2024-06-27/spacex-tender-offer-said-to-value-company-at-record-210-billion>.

Roush, T. (2024). Elon Musk's X Is Now Worth Around A Fifth Of The \$44 Billion He Paid For It, Fidelity Says. *Forbes*. [online] 30 Sep. Available at: <https://www.forbes.com/sites/tylerroush/2024/09/30/elon-musks-x-is-now-worth-around-a-fifth-of-the-44-billion-he-paid-for-it-fidelity-says/> [Accessed 28 Oct. 2024].

Saltz Mongeluzzi Bendesky P.C. (2024). *Saltz Mongeluzzi Bendesky P.C.* [online] Available at: <https://www.smbb.com/news-article/devastating-tiktok-lawsuit-may-have-killed-section-230-is-social-media-doomed/>.

Sanger, D.E. and Lipton, E. (2023). The White House May Condemn Musk, but the Government Is Addicted to Him. *The New York Times*. [online] 19 Nov. Available at: <https://www.nytimes.com/2023/11/19/us/politics/elon-musk-white-house-pentagon.html?smid=nytcore-ios-share&referringSource=articleShare&sgrp=c-cb> [Accessed 28 Oct. 2024].

Satariano, A., Metz, C. and Mickle, T. (2024). Elon Musk Sues OpenAI and Sam Altman for Violating the Company's Principles. *The New York Times*. [online] 1 Mar. Available at: <https://www.nytimes.com/2024/03/01/technology/elon-musk-openai-sam-altman-lawsuit.html>.

Sauer, M. (2022). *Mark Zuckerberg doesn't like your scrolling habits: Social media is for 'building relationships,' not just consuming content*. [online] CNBC. Available at:

<https://www.cnn.com/2022/09/04/zuckerberg-social-media-is-for-building-relationships-not-scrolling.html>.

Schleifer, T. and Isaac, M. (2024). Mark Zuckerberg's Political Evolution. *The New York Times*. [online] 24 Sep. Available at: <https://www.nytimes.com/2024/09/24/technology/mark-zuckerberg-trump-politics.html?smid=nytcore-ios-share&referringSource=articleShare&sgrp=c-cb> [Accessed 28 Oct. 2024].

Section 230: A Brief Overview. (2024). Available at: <https://crsreports.congress.gov/product/pdf/IF/IF12584>.

Shaban, H. and Siddiqui, F. (2022). *Here's who helped Elon Musk buy Twitter*. [online] Washington Post. Available at: <https://www.washingtonpost.com/technology/2022/12/24/elon-musk-twitter-funders/>.

Shoemaker, J.D. (2023). *Tesla's solar factory in Buffalo fizzles - Investigative Post*. [online] Investigative Post. Available at: <https://www.investigativepost.org/2023/01/11/teslas-solar-factory-in-buffalo-fizzles/> [Accessed 28 Oct. 2024].

Siddiqui, F. (2022). Elon Musk found not liable in trial over Tesla's SolarCity acquisition. *Washington Post*. [online] 27 Apr. Available at: <https://www.washingtonpost.com/technology/2022/04/27/musk-solarcity-verdict/>.

Soni, J. (2023). *The Colorful History of X.com (Aka the website formerly known as Twitter)*. [online] Medium. Available at: <https://medium.com/@jimmysoni/the-colorful-history-of-x-com-aka-the-website-formerly-known-as-twitter-53b3dfec0069>.

Sottile, R.R. (2023). *Uber self-driving car test driver pleads guilty to endangerment in pedestrian death case | CNN Business*. [online] CNN. Available at: <https://www.cnn.com/2023/07/29/business/uber-self-driving-car-death-guilty/index.html>.

SpaceX (2024). *Mission*. [online] SpaceX. Available at: <https://www.spacex.com/mission/>.

SpaceX Eyes \$15 Billion in 2024 Sales on Starlink Strength. (2023). *Bloomberg.com*. [online] 6 Nov. Available at: <https://www.bloomberg.com/news/articles/2023-11-06/spacex-eyes-15-billion-in-sales-next-year-on-starlink-strength>.

Spangler, T. (2023). *White House Slams Elon Musk's 'Abhorrent Promotion of Antisemitic and Racist Hate'*. [online] Variety. Available at: <https://variety.com/2023/digital/news/white-house-condemns-elon-musk-antisemitic-racist-hate-1235795049/>.

Sprout Social (2024). *50+ of the Most Important Social Media Marketing Statistics for 2023*. [online] Sprout Social. Available at: <https://sproutsocial.com/insights/social-media-statistics/>.

Su, J. (2018). Tesla Could Have Full Self-Driving Cars On The Road By 2019, Elon Musk Says. *Forbes*. [online] 7 Nov. Available at: <https://www.forbes.com/sites/jeanbaptiste/2018/11/07/tesla-could-have-full-self-driving-cars-on-the-road-by-2019-elon-musk-says/> [Accessed 28 Oct. 2024].

Tesla (2024a). *Elon Musk | Tesla*. [online] Tesla.com. Available at: <https://www.tesla.com/elon-musk>.

Tesla. (2024b). *Master Plan, Part Deux | Tesla*. [online] Available at: <https://www.tesla.com/master-plan-part-deux>.

Teslam (2024). *Tesla vs Waymo: The Battle for Autonomous Vehicle Leadership*. [online] Tesla Mag. Available at: <https://www.tesla-mag.com/en/tesla-vs-waymo-the-battle-for-autonomous-vehicle-leadership/>.

The Economist. (2018). *Tesla is heading for a cash crunch*. [online] Available at: <https://www.economist.com/business/2018/04/05/tesla-is-heading-for-a-cash-crunch>.

The Economist. (2024). *Elon Musk is feuding with Brazil's powerful Supreme Court*. [online] Available at: https://www.economist.com/the-americas/2024/04/14/elon-musk-is-feuding-with-brazils-powerful-supreme-court?utm_medium=cpc.adword.pd&utm_source=google&ppccampaignID=17210591673&ppcadID=&utm_campaign=a.22brand_pmax&utm_content=conversion.direct-response.anonymous&gad_source=1&gclid=EAIaIQobChMI4qeDuruxiQMVSjUIBR3WSyOpEAAAYASAAEgIBkFD_BwE&gclidsrc=aw.ds.

The Tesla Team (2023). *Master Plan Part 3*. [online] Tesla. Available at: <https://www.tesla.com/blog/master-plan-part-3>.

Thompson, C. (2016). *Elon Musk: Fully autonomous Tesla will drive across the country by the end of 2017*. [online] Business Insider. Available at: <https://www.businessinsider.com/elon-musk-autonomous-tesla-drive-across-country-by-end-of-2017-2016-10> [Accessed 28 Oct. 2024].

Twitter. (n.d.). <https://twitter.com/elonmusk/status/1525049369552048129>. [online] Available at: <https://twitter.com/elonmusk/status/1525049369552048129>.

Waymo (2023). *Waymo*. [online] Waymo. Available at: <https://waymo.com/>.

Wile, R. (2022). *A Timeline of Elon Musk's Takeover of Twitter*. [online] NBC News. Available at: <https://www.nbcnews.com/business/business-news/twitter-elon-musk-timeline-what-happened-so-far-rcna57532>.

WSJ. (n.d.). *How Elon Musk Made His Fortune*. [online] Available at: <https://www.wsj.com/story/how-elon-musk-made-the-fortune-hell-need-to-buy-twitter-c9e15543>.

www.cbsnews.com. (2023a). *Elon Musk says he wants to create his own 'safer' version of ChatGPT called TruthGPT*. [online] Available at: <https://www.cbsnews.com/news/elon-musk-artificial-intelligence-chatgpt-truthgpt-safer-ai-tucker-carlson-fox-news-interview/>.

www.cbsnews.com. (2023b). *Tesla's Autopilot not responsible for fatal 2019 crash in California, jury finds in landmark case - CBS News*. [online] Available at: <https://www.cbsnews.com/news/tesla-autopilot-not-responsible-fatal-2019-crash-california-jury-finds-landmark-case/>.

www.encyclopedia.com. (n.d.). *X.com | Encyclopedia.com*. [online] Available at: <https://www.encyclopedia.com/economics/encyclopedias-almanacs-transcripts-and-maps/xcom>.

www.linkedin.com. (n.d.). *SpaceX's Launch to Success: How Federal Grants and Contracts Helped Elon Musk Reach New Heights*. [online] Available at: <https://www.linkedin.com/pulse/spacexs-launch-success-how-federal-grants-contracts-new-emdjian-mba>.

www.mentorist.app. (n.d.). *The Founders: The Story of Paypal and the Entrepreneurs Who Shaped Silicon Valley - Mentorist app*. [online] Available at: <https://www.mentorist.app/books/the-founders-the-story-of-paypal-and-the-entrepreneurs-who-shaped-silicon-valley/>.

www.reed.senate.gov. (2023). *Sen. Reed: Starlink Highlights Vulnerabilities That Must be Addressed | U.S. Senator Jack Reed of Rhode Island*. [online] Available at: <https://www.reed.senate.gov/news/releases/sen-reed-starlink-highlights-vulnerabilities-that-must-be-addressed>.

www.sec.gov. (n.d.). *SEC.gov | SEC v. Elon Musk Case No. 18-cv-8865 (S.D.N.Y.) SEC v. Tesla, Inc. Case No. 18-cv-8947 (S.D.N.Y.)*. [online] Available at: <https://www.sec.gov/enforcement-litigation/distributions-for-harmed-investors/sec-v-elon-musk-case-no-18-cv-8865-sdny-sec-v-tesla-inc-case-no-18-cv-8947-sdny>.

X (formerly Twitter). (2024a). *x.com*. [online] Available at: <https://x.com/elonmusk/status/1026872652290379776?lang=en>.

X (formerly Twitter). (2024b). *x.com*. [online] Available at: <https://x.com/elonmusk/status/984705630106673152?prefetchTimestamp=1727752142882> [Accessed 28 Oct. 2024].

x.ai. (n.d.). *About xAI*. [online] Available at: <https://x.ai/about>.

Zahn, M. (2022). *A timeline of Elon Musk's tumultuous Twitter acquisition attempt*. [online] ABC News. Available at: <https://abcnews.go.com/Business/timeline-elon-musks-tumultuous-twitter-acquisition-attempt/story?id=86611191>.